COUNTING ERDŐS–STRAUS SOLUTIONS ON ONE TYPE II LINE: A POINTWISE BOUND BELOW $n^{1/3}$, AN UNCONDITIONAL GENUS MAIN TERM, AND $(\log p)^2$ FOR ALMOST ALL PRIMES

ANONYMOUS

Abstract. For an integer $n \geq 2$ let $R(n)$ be the number of triples (u, B, C) of positive integers with $4uBC - u - B = n$. Each such triple gives the solution $4/n = 1/(uBC) + 1/(nBC) + 1/(nuC)$ of the Erdős–Straus equation; for prime n these form one line inside the Type II solutions of Elsholtz and Tao. We prove three estimates for R. (i) For every n, $R(n) \ll_\varepsilon n^{12/37+\varepsilon}$, a power saving over the divisor-bound benchmark $n^{1/3+o(1)}$. The proof applies Heath-Brown's square sieve to the quadratic $(4Ck - n)^2 - 4k$, whose square values are exactly the solutions, followed by Poisson summation, Weil's bound for Kloosterman sums and van der Corput's third-derivative test. A cruder version of the same argument, using only complete quadratic character sums, gives $n^{1/3-1/225}(\log n)^2$. Any exponent pair $(\kappa, \kappa + \frac{1}{2})$ with $\kappa < \frac{1}{4}$ gives $n^{4/(13-4\kappa)+\varepsilon}$, so the exponent is $21/65$ relative to Bourgain's exponent pair and $4/13$ under the exponent pair conjecture. (ii) For primes $p \equiv 1 \pmod 4$, the part $M_{\mathrm{gen}}(p)$ of the natural expected count of solutions that comes from real Dirichlet characters satisfies $M_{\mathrm{gen}}(p) \geq (2/(3\pi^4) - o(1))(\log p)^2$ for every sufficiently large p, unconditionally and effectively. The proof combines Burgess's bound for cube-free conductors with a windowed hyperbola method in which all non-principal genus characters cancel. We also give an asymptotic formula for the genus layer of each short slice in terms of products $L(1, \psi)L(1, \psi')$ of quadratic L-values, and show that on the same subfamily the characters of order at least three modulo the small modulus carry no density main term. (iii) For all but $O_\eta(\pi(X)/\log X)$ primes $p \leq X$ we have $R(p) \geq (1 - \eta)(\log p)^2/64$. This is a variance bound over primes by Bombieri–Vinogradov for a truncated count, in which every residue class becomes a single hyperbolic lattice cell. It improves the almost-all exponent 0.549 of Elsholtz–Tao to 2 for the solution count. The Erdős–Straus conjecture remains open: (i) is an upper bound and holds at odd squares, where $R = 0$; (ii) concerns a main term and not $R(p)$ itself; (iii) says nothing about any prescribed prime.

1. Introduction

1.1. **The equation and the count.** The Erdős–Straus conjecture asserts that for every integer $n \geq 2$

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \tag{1}$$

has a solution in positive integers. It has been verified for $n \leq 10^{17}$ [15]. A solution for n gives one for every multiple of n, so it suffices to treat primes. The classes $n \equiv 0, 2, 3 \pmod 4$ have polynomial solutions, and Mordell's identities [14] leave only primes in the six classes $1, 121, 169, 289, 361, 529 \pmod{840}$. Vaughan [20] showed that the number of exceptions up to N is $\ll N \exp(-c(\log N)^{2/3})$. The conjecture is open, and nothing in this paper proves it for any new integer.

Elsholtz and Tao [8] call a solution of (1) *Type I* if $n \mid x$ and n is coprime to y, z, and *Type II* if $n \mid y, z$ and n is coprime to x. They write $f(n)$, $f_{\mathrm{I}}(n)$, $f_{\mathrm{II}}(n)$ for the numbers of ordered

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solutions of each kind, and show [8, (1.4)] that $f(p) = 3f_{\mathrm{I}}(p) + 3f_{\mathrm{II}}(p)$ for odd primes p. We study the following one-parameter subfamily of Type II solutions. For $n \geq 2$ let

$$R(n) := \#\{(u, B, C) \in \mathbb{N}^3 : \ 4uBC - u - B = n\}. \tag{2}$$

A triple counted by $R(n)$ gives the solution

$$\frac{4}{n} = \frac{1}{uBC} + \frac{1}{nBC} + \frac{1}{nuC}. \tag{3}$$

In Elsholtz–Tao's parametrisation of Type II solutions [8, Prop. 2.6, (2.13)–(2.22)] this is the slice $c = 1$, with $(a, b, d) = (B, u, C)$ and $e = u + B$. The solution (3) is of Type II when $\gcd(n, uBC) = 1$, in particular for every prime $n = p$ (but not in general: $n = 6$, $(u, B, C) = (1, 1, 2)$ gives $(2, 12, 12)$). For a prime p the map from triples to solutions is injective, so $f_{\mathrm{II}}(p) \geq R(p)$ (Lemma 2.1). The count $R(p)$ is the natural "first line" of Type II solutions: it is a sum of $\asymp p \log p$ rare divisibility events, each tied to its own modulus, whose expected total is of order $(\log p)^2$ (§1.3). It vanishes at some primes: among primes $p \equiv 1 \pmod 4$ below 10^7 we find $R(p) = 0$ exactly for

$$409,\ 577,\ 5569,\ 9601,\ 23929,\ 83449,\ 102001,$$
$$329617,\ 712321,\ 1134241,\ 1724209,\ 1726201,\ 5212561,\ 8813281$$

(a direct computation; §6). These primes have other solutions of (1), so $R(p) \geq 1$ is strictly stronger than the conjecture at p. Nothing below shows $R(p) \geq 1$ for any prescribed prime.

Two classical facts frame everything that follows.

(1) *Odd squares.* $R(m^2) = 0$ for every odd m (Lemma 2.3). This is the analogue for our family of Elsholtz–Tao's Proposition 1.6 (odd squares have no Type I or Type II solutions), which goes back to Schinzel and Yamamoto [22]; see also Bright–Loughran [3, Cor. 1.4]. Hence every argument that proves a positive lower bound for $R(n)$ must contain a step that is false at odd squares. Upper bounds and averages over n are not subject to this constraint, and this is why they are the kind of result we can prove.

(2) *The divisor bound.* Fixing the smallest of u, B, C reduces $R(n)$ to divisor counts of single integers of size $n^{O(1)}$, which gives $R(n) \ll n^{1/3+o(1)}$ (Lemma 2.4). Elsholtz and Tao prove $f_{\mathrm{II}}(n) \ll n^{2/5+o(1)}$ for the full Type II count [8, Prop. 1.7] and remark that this "appears to be the limit of what one can obtain purely from the divisor bound". Their bound concerns a larger family and is not comparable with ours.

1.2. **An upper bound below the divisor bound.**

Theorem A. *For every $\varepsilon > 0$ and every integer $n \geq 2$,*

$$R(n) \ll_\varepsilon n^{12/37+\varepsilon} \qquad (12/37 = 0.3243\ldots).$$

Moreover:

(a) *$R(n) \ll n^{1/3-1/225}(\log n)^2$ for $n \geq 2$, by an argument that uses only the evaluation of complete quadratic character sums (no Kloosterman sums), Vaaler's lemma, the third-derivative test and Shiu's theorem (Theorem 3.8).*

(b) *If $(\kappa, \kappa + \frac{1}{2})$ is an exponent pair with $0 \leq \kappa < \frac{1}{4}$, then $R(n) \ll_\varepsilon n^{4/(13-4\kappa)+\varepsilon}$ (Theorem 3.14). The pair $(\frac{1}{6}, \frac{2}{3})$ recovers $12/37$. Bourgain's pair $(\frac{13}{84} + \varepsilon, \frac{55}{84} + \varepsilon)$ [2, Thm. 6] gives $R(n) \ll_\varepsilon n^{21/65+\varepsilon}$, relative to that theorem (see Remark 3.15). The exponent pair conjecture, that $(\varepsilon, \frac{1}{2} + \varepsilon)$ is an exponent pair, would give $R(n) \ll_\varepsilon n^{4/13+\varepsilon}$.*

The key observation is that near the "cube" $u, B, C \asymp n^{1/3}$, which carries the divisor-bound benchmark, a solution is a point (C, k), $k = uB$, of a thin strip at which the quadratic

$$f(C, k) = (4Ck - n)^2 - 4k$$

is a perfect square: u and B are the roots of $z^2 - (4Ck - n)z + k$. Heath-Brown's square sieve [11] turns this into character sums $\sum \left(\frac{f(C,k)}{qq'}\right)$ over the strip. Applying Poisson summation in k inside each fixed C turns the character into complete Kloosterman sums times exponential sums in C, to which the third-derivative test applies. The argument uses no property of n and holds verbatim at odd squares, where $R(n) = 0$. It is an upper bound only, and does not bear on the existence of solutions.

1.3. **The genus main term at every large prime.** For an odd prime p write $\mathcal{P}(p) = \{(u, B) \in \mathbb{N}^2 : \gcd(u, B) = 1,\ u+B \text{ odd},\ 4uB-u-B \le p\}$, and for $(u, B) \in \mathcal{P}(p)$ put $q = 4uB$, $a = -(u + B)$; then a and p are units modulo q and $R(p) = \sum_{(u,B)\in\mathcal{P}(p)} \mathbf{1}[p \equiv a \pmod q]$ (Lemma 2.2). Expanding each indicator into Dirichlet characters modulo q and keeping only the real characters gives

$$M_{\mathrm{gen}}(p) := \sum_{(u,B)\in\mathcal{P}(p)} \frac{s(q)}{\varphi(q)}\, \mathbf{1}[pa \in ((\mathbb{Z}/q\mathbb{Z})^\times)^2], \qquad s(q) := \big|(\mathbb{Z}/q\mathbb{Z})^\times/((\mathbb{Z}/q\mathbb{Z})^\times)^2\big|, \tag{4}$$

so that $R(p) = M_{\mathrm{gen}}(p) + X(p)$, where $X(p)$ is the contribution of the characters of order at least 3. Every term of M_{gen} is nonnegative, $M_{\mathrm{gen}}(p)$ depends on p only through quadratic data, and the same expression with p replaced by an odd square vanishes identically (Remark 4.1). It is the "genus" expected count.

Theorem B. *For every $\varepsilon > 0$ there is an effectively computable $p_0(\varepsilon)$ such that every prime $p \equiv 1 \pmod 4$ with $p \ge p_0(\varepsilon)$ satisfies*

$$M_{\mathrm{gen}}(p) \ \ge\ \Big(\frac{2}{3\pi^4} - \varepsilon\Big)(\log p)^2.$$

The proof (§4) uses a subfamily of pairs $(u, 2w), (2w, u)$ with u and w in sharp windows. On it the genus condition factors into a character in w of conductor dividing $4u$ times the condition $\chi_{-4up}(\ell) = 1$ at each prime $\ell \mid w$. The window *difference* cancels every constant term of the hyperbola method, so of the $2^{\omega(u)+1}$ genus characters only the principal one survives. This cancellation needs Burgess's bound for characters of cube-free conductor, and works because p divides the conductor of each twisted character $\eta\chi_{-4up}$. An average of $L(1, \chi_{-4up})$ over u, again by Burgess, finishes the proof. No zero-free region, no Siegel-type bound and no averaging over p is used. Section 4 also contains two complements: a uniform asymptotic formula, with power-saving additive error, for the genus layer of each short slice $u \equiv 1 \pmod 4$ squarefree, $u \le p^{3/5-\varepsilon}$, as a principal term (involving $L(1, \chi_{-4up})$ and $L'(1, \chi_{-4up})$) plus a signed sum of products of two quadratic L-values (Proposition 4.10), and a density-level statement that the characters of order ≥ 3 modulo $8u$ carry no main term on the subfamily (Theorem 4.12).

Theorem B is a statement about M_{gen}, not about R. It shows that no configuration of quadratic characters can make the expected count small at a large prime. It gives no information on the remainder $X(p)$, and $R(p) \ge 1$ does not follow from it.

1.4. **Almost all primes.**

Theorem C. *There is an absolute constant c_0 such that for every $\eta \in (0, 1)$ and $X \ge X_0(\eta)$*

$$\#\Big\{p \le X \text{ prime}:\ R(p) < (1-\eta)\frac{(\log p)^2}{64}\Big\} \ \le\ \frac{c_0}{\eta^2}\cdot\frac{\pi(X)}{\log X};$$

$c_0 = 1600$ is admissible. The threshold $X_0(\eta)$ is ineffective. In particular $f(p) \geq 3f_{\mathrm{II}}(p) \geq 3R(p) \gg (\log p)^2$ for all primes outside a set of relative density zero.

Elsholtz and Tao prove $f(p) \geq (\log p)^{(\log 3)/2-o(1)}$ with $(\log 3)/2 = 0.549\ldots$ for all primes in a set of relative density one [8, Thm. 1.8], and remark that higher moments of $f_{\mathrm{I}}(p)$, $f_{\mathrm{II}}(p)$ "seem ... out of reach of our methods, as the level of the relevant divisor sums becomes too great to handle" [8, Rem. 1.3]. Theorem C bypasses the second moment of R itself. It keeps only the solutions certified by moduli $4uC - 1 \leq X^{1/4-o(1)}$, so that all pair correlations fall within the range of Bombieri–Vinogradov. In the resulting variance each residue class modulo d is a single lattice cell of a hyperbolic region, which makes it $O(\log^3 X)$ against a squared mean of order $\log^4 X$. The order $(\log p)^2$ is the natural order of R: the number of triples with $4uBC - u - B \leq X$ lies between $\#\{uBC \leq X/4\}$ and $\#\{uBC \leq X/2\}$, so the average of $R(n)$ over $n \leq X$ is $\asymp (\log X)^2$. By contrast, the mean of $f_{\mathrm{II}}(p)$ over primes is of order $(\log p)^3$ [8, Thm. 1.1].

1.5. **What is and is not proved.** All four theorems are proved here in full, relative to the cited classical inputs (Weil's bound, the third-derivative test, exponent pairs where stated, Burgess's bound, Bombieri–Vinogradov, Brun–Titchmarsh-type and divisor-sum bounds). They are not externally refereed. We state their limits explicitly.

- The Erdős–Straus conjecture is *not* proved; nor is $R(p) \geq 1$ for all large p, nor for any prescribed prime.
- Theorem A is an upper bound uniform in n; it is true at odd squares, where $R = 0$. It improves only the size of the fluctuations that $R(p)$ could have.
- Theorem B bounds the real-character part $M_{\mathrm{gen}}(p)$ of the expected count. The remainder $X(p) = R(p) - M_{\mathrm{gen}}(p)$ is not controlled: a lower bound $X(p) \geq -(1-\varepsilon)M_{\mathrm{gen}}(p)$ for all large p would, with Theorem B, prove the conjecture for all large primes, and we have no approach to it. Theorem 4.12 controls only a density version of part of X.
- Theorem C is an average over p; the fourteen primes listed above are compatible with it.

1.6. **Literature.** A search of the arXiv and the web (queries on Erdős–Straus solution counts, Type II counts, square sieve applications, and divisor counts in residue classes; details in the accompanying review file) found no earlier pointwise bound for $R(n)$ or for any Type II family below $n^{1/3}$, no almost-all lower bound of order $(\log p)^2$ for primes, and no unconditional lower bound for a genus main term of this kind. We make no stronger priority claim. The pointwise bounds we know of are Elsholtz–Tao's $f_{\mathrm{I}}(n) \ll n^{3/5+o(1)}$, $f_{\mathrm{II}}(n) \ll n^{2/5+o(1)}$ [8, Prop. 1.7] and the bound $O_\varepsilon(n^\varepsilon(n^3/m^2)^{1/5})$ of Elsholtz–Planitzer [7, Thm. 1] for representations of m/n. The square sieve was introduced in [11]; the combination of a power sieve with a q-analogue of van der Corput's method has been used to count rational points on cyclic covers of projective space [12]. The variance-over-primes argument of Theorem C is a Turán–Kubilius-type estimate for a non-additive divisor count; the closest classical analogue we know is Barban's inequality for shifted primes, used by Elsholtz–Tao in the proof of their Theorem 1.8.

1.7. **Notation.** $\mathbb{N} = \{1, 2, \ldots\}$. $e(x) = e^{2\pi i x}$. $\left(\frac{a}{b}\right)$ is the Jacobi symbol for odd $b > 0$ (Legendre for prime b; for $a < 0$ it is defined through the residue of a). For a discriminant D, $\chi_D = \left(\frac{D}{\cdot}\right)$ is the Kronecker symbol, and χ_{-4} is the non-principal character modulo 4. τ, ω, φ, μ are the divisor function, the number of distinct prime factors, Euler's function and the Möbius function. $\psi(x) = x - \lfloor x \rfloor - \frac{1}{2}$ and $\log^+ x = \max(\log x, 0)$. Implied constants are absolute unless indicated by subscripts; $n^{o(1)}$ absorbs factors such as $\tau(m)$ for $m \leq n^{O(1)}$.

2. Preliminaries on $R(n)$

Lemma 2.1. *Let $n \geq 2$.*

(i) *Every triple (u, B, C) counted by $R(n)$ satisfies $n < 4uBC \leq 2n$; in particular $u, B, C \leq n/2$.*
(ii) *$(u, B, C) \mapsto (B, u, C)$ is an involution of the solution set.*
(iii) *If $n = p$ is prime, (3) maps the solution set injectively into the Type II solutions of $4/p$; hence $f_{\mathrm{II}}(p) \geq R(p)$.*

Proof. (i) $u + B \leq 2uB \leq 2uBC$, so $n = 4uBC - (u + B) \geq 2uBC$; and $n < 4uBC$. (ii) is clear. (iii) Multiplying (3) by $nuBC$ turns it into $4uBC = n + u + B$, so (3) holds. By (i), $u, B, C < p$, so $p \nmid uBC$ while p divides pBC and puC: the solution is of Type II. From $(x, y, z) = (uBC, pBC, puC)$ we recover $u = px/y$, $B = px/z$ and $C = x/(uB)$. □

Lemma 2.2 (pair form)**.** *Let p be an odd prime. The map $(u, B, C) \mapsto (u, B)$ is a bijection from the triples counted by $R(p)$ onto the pairs $(u, B) \in \mathcal{P}(p)$ with $p \equiv a \pmod q$, where $q = 4uB$ and $a = -(u + B)$. For $(u, B) \in \mathcal{P}(p)$, both a and p are units modulo q.*

Proof. Given a triple, $4uB - u - B \leq 4uBC - u - B = p$; $u + B \equiv p \equiv 1 \pmod 2$; and $\gcd(u, B)$ divides $4uBC - u - B = p$ and is at most $u < p$, so it is 1. Clearly $p \equiv a \pmod q$. Conversely, if $(u, B) \in \mathcal{P}(p)$ and $q \mid p - a = p + u + B$ then $C := (p + u + B)/(4uB)$ is a positive integer. Finally, for $(u, B) \in \mathcal{P}(p)$, $\gcd(u + B, 4uB) = 1$ because $u + B$ is odd and $\gcd(u + B, u) = \gcd(B, u) = 1$; and $p \nmid q$ because $p \geq 4uB - u - B \geq 2uB$ gives $u, B \leq p/2$. □

Lemma 2.3 (odd squares)**.** *Let $n \geq 3$ be odd and let (u, B, C) be a triple counted by $R(n)$. Put $x = 4Cu - 1$, $y = 4CB - 1$. Then $xy = 4Cn + 1$ and $\left(\frac{n}{x}\right) = -1$. In particular $R(m^2) = 0$ for every odd $m \geq 3$. If moreover $\gcd(uBC, n) = 1$, then $\left(\frac{-uB}{n}\right) = -1$.*

Proof. $xy = 16C^2uB - 4C(u+B)+1 = 4C(4uBC - u - B)+1 = 4Cn+1$. Hence $\gcd(x, n) = 1$ and $4Cn \equiv -1 \pmod x$. As $x \equiv 3 \pmod 4$, $\left(\frac{4Cn}{x}\right) = \left(\frac{-1}{x}\right) = -1$. Write $C = 2^aC'$ with C' odd. If $a \geq 1$ then $8 \mid 4C \mid x + 1$, so $\left(\frac{2}{x}\right) = 1$. By reciprocity $\left(\frac{C'}{x}\right) = \left(\frac{x}{C'}\right)(-1)^{\frac{C'-1}{2}\cdot\frac{x-1}{2}}$; since $x \equiv -1 \pmod{C'}$ and $\frac{x-1}{2}$ is odd, this equals $(-1)^{\frac{C'-1}{2}}(-1)^{\frac{C'-1}{2}} = 1$. So $\left(\frac{C}{x}\right) = 1$ and $\left(\frac{n}{x}\right) = -1$. For $n = m^2$ this contradicts $\left(\frac{m^2}{x}\right) = 1$. For the last claim, reciprocity gives $\left(\frac{x}{n}\right) = \left(\frac{n}{x}\right)(-1)^{\frac{n-1}{2}\cdot\frac{x-1}{2}} = -\left(\frac{-1}{n}\right)$. Modulo n we have $xy \equiv 1$, hence $x(x+1)(y+1) \equiv (x+1)^2$, and $\gcd(x + 1, n) = \gcd(4Cu, n) = 1$, so $\left(\frac{x}{n}\right) = \left(\frac{(x+1)(y+1)}{n}\right) = \left(\frac{16C^2uB}{n}\right) = \left(\frac{uB}{n}\right)$. Therefore $\left(\frac{-uB}{n}\right) = \left(\frac{-1}{n}\right)\left(\frac{x}{n}\right) = -1$. □

Lemma 2.4 (divisor-bound benchmark)**.** *Let $\mathfrak{B}(U, V, W)$ be the set of triples counted by $R(n)$ with $u \in (U, 2U]$, $B \in (V, 2V]$, $C \in (W, 2W]$, and $N(U, V, W)$ its size. Then $N(U, V, W) \ll_\varepsilon n^\varepsilon \min(U, V, W)$, and $R(n) \ll_\varepsilon n^{1/3+\varepsilon}$.*

Proof. For fixed u, $B(4uC - 1) = n + u$, so B is a divisor of $n + u$ and C is determined by B. For fixed C, $x = 4Cu - 1$ is a divisor of $4Cn + 1 \leq 2n^2 + 1$ and determines u and then B. The case of fixed B is symmetric. The divisor bound gives the first claim. By Lemma 2.1(i), $\min(u, B, C) \leq (n/2)^{1/3}$; fixing the smallest coordinate in the same way gives the second. □

Dyadic blocks will be indexed by $U, V, W \in \{2^j/2 : j \geq 0\}$, so that every triple lies in exactly one block. By Lemma 2.1(i), nonempty blocks satisfy $n/32 < UVW < n/2$, and there are $O((\log n)^3)$ of them.

3. The upper bound: square detection on a strip

3.1. **Square detection.** For integers C, k put

$$f(C,k) := (4Ck-n)^2 - 4k = 16C^2k^2 - (8Cn+4)k + n^2. \tag{5}$$

Lemma 3.1. *Let (u,B,C) be a triple counted by $R(n)$, and put $k = uB$, $s = u+B$. Then $4Ck - n = s$ and $f(C,k) = (u-B)^2$. Conversely, if $C \geq 1$, $4Ck - n > 0$ and $f(C,k) = r^2$ with $r \in \mathbb{Z}_{\geq 0}$, then $u = (s-r)/2$, $B = (s+r)/2$ with $s = 4Ck - n$ are positive integers with $4uBC - u - B = n$, $u + B = s$, $uB = k$. In particular every (C,k) is the image of at most two triples, namely (u,B,C) and (B,u,C). Triples with $u = B$ number at most $\tau(n)$.*

Proof. $4Ck - s = 4uBC - u - B = n$ and $f = s^2 - 4uB = (u-B)^2$. Conversely $4Ck = n + s > 0$ gives $k \geq 1$; $s^2 - r^2 = 4k > 0$ with $s > 0$ gives $0 \leq r < s$ and $s \equiv r \pmod 2$, so $u \geq 1$; then $u + B = s$, $uB = (s^2 - r^2)/4 = k$ and $4uBC - u - B = 4Ck - s = n$. Since $\{u, B\}$ is the root set of $z^2 - sz + k$, the preimages are as stated. If $u = B$ then $u(4uC - 2) = n$, so $u \mid n$, and u determines C. □

3.2. **The square sieve.** We use the following variant of Heath-Brown's square sieve [11, Thm. 1], with explicit constants and a support condition on positive squares only. (Heath-Brown's theorem assumes $w(a) = 0$ for $|a| \geq e^{P_0}$; the proof below is his.)

Lemma 3.2. *Let $\mathcal{P}$ be a set of $P_0 \geq 1$ odd primes and $w : \mathbb{Z} \setminus \{0\} \to [0,\infty)$ finitely supported, with $w(a) = 0$ for every perfect square $a > 2^{P_0}$. Then*

$$\sum_{m \geq 1} w(m^2) \;\leq\; \frac{4}{P_0} \sum_a w(a) + \frac{4}{P_0^2} \sum_{\substack{q,q' \in \mathcal{P} \\ q \neq q'}} \Big| \sum_a w(a) \left(\frac{a}{qq'}\right) \Big|.$$

Proof. Let $a = m^2$ with $1 \leq m \leq 2^{P_0/2}$. Then m has at most $\log_2 m \leq P_0/2$ prime factors, so $\sigma(a) := \sum_{q \in \mathcal{P}} \left(\frac{a}{q}\right) = \#\{q \in \mathcal{P} : q \nmid m\} \geq P_0/2$. Hence $\sum_m w(m^2) \leq \frac{4}{P_0^2} \sum_a w(a)\sigma(a)^2$ (the added terms are ≥ 0). Expanding, $\sigma(a)^2 = \sum_q \left(\frac{a}{q}\right)^2 + \sum_{q \neq q'} \left(\frac{a}{qq'}\right) \leq P_0 + \sum_{q \neq q'} \left(\frac{a}{qq'}\right)$. □

3.3. **Complete character sums.** In this section q, q' denote odd primes and $\ell \in \mathbb{Z}$ is a frequency. For an odd prime q and $c, \ell \in \mathbb{Z}$ put

$$K_q(c) := \sum_{k \bmod q} \left(\frac{f(c,k)}{q}\right), \qquad G_c^{(q)}(\ell) := \sum_{k \bmod q} \left(\frac{f(c,k)}{q}\right) e\Big(\frac{\ell k}{q}\Big),$$

and for $Q = qq'$ with distinct odd primes q, q' let $G_c^{(Q)}(\ell) := \sum_{k \bmod Q} \left(\frac{f(c,k)}{Q}\right) e(\ell k/Q)$. Let $D_q := \{c \in \mathbb{Z} : q \mid 4cn + 1\}$, a single residue class modulo q (empty if $q \mid n$).

Cited theorem 3.3 (Weil; [13, Thm. 11.11, Cor. 11.12])**.** For a prime q and integers a, b with $q \nmid ab$, the Kloosterman sum $S(a,b;q) = \sum_{x \bmod q,\, x \neq 0} e((ax + b\bar{x})/q)$ satisfies $|S(a,b;q)| \leq 2\sqrt{q}$.

Lemma 3.4. *Let q be an odd prime and $c, \ell \in \mathbb{Z}$.*

(a) *$K_q(c) = 0$ if $q \mid c$; $K_q(c) = q - 1$ if $c \in D_q$ (then $q \nmid c$); and $K_q(c) = -1$ otherwise. In particular $|K_q(c)| \leq 1 + q\mathbf{1}_{D_q}(c)$.*

(b) *If $q \nmid \ell$ then $|G_c^{(q)}(\ell)| \leq 2\sqrt{q}$.*

(c) *$\sum_{c \bmod q} |G_c^{(q)}(\ell)| \leq 2q^{3/2}$ for every ℓ.*

(d) *For $Q = qq'$, $G_c^{(Q)}(\ell) = G_c^{(q)}(\ell\bar{q}')\, G_c^{(q')}(\ell\bar{q})$, where $q'\bar{q}' \equiv 1 \pmod q$ and $q\bar{q} \equiv 1 \pmod{q'}$. Hence $\sum_{c \bmod Q} |G_c^{(Q)}(\ell)| \leq 4Q^{3/2}$ for every ℓ, and $K(c) := G_c^{(Q)}(0) = K_q(c)K_{q'}(c)$.*

Proof. (a) If $q \mid c$ then $f(c,k) \equiv n^2 - 4k$ runs over all residues once, so the sum is $\sum_t \left(\frac{t}{q}\right) = 0$. If $q \nmid c$, $f(c,\cdot)$ is a quadratic polynomial with leading coefficient $16c^2$, a nonzero square, and discriminant $(8cn+4)^2 - 64c^2n^2 = 16(4cn+1)$. Completing the square, $f(c,k) \equiv 16c^2((k-r)^2 - \delta)$ with $\delta \equiv 0$ iff $c \in D_q$, so $K_q(c) = \sum_x \left(\frac{x^2-\delta}{q}\right)$. If $\delta \equiv 0$ this is $q-1$. If $\delta \not\equiv 0$ it equals $\sum_x (\#\{y : y^2 \equiv x^2 - \delta\} - 1) = \#\{(x,y) : (x-y)(x+y) \equiv \delta\} - q = (q-1) - q = -1$.

(b) If $q \mid c$, substitute $t = n^2 - 4k$: $G = e(\bar{4}\ell n^2/q) \sum_t \left(\frac{t}{q}\right) e(-\bar{4}\ell t/q)$, a Gauss sum of modulus $\sqrt{q}$. If $q \nmid c$, put $A = 16c^2$, $r \equiv (8cn+4)(2A)^{-1}$ and $d \equiv 16(4cn+1)(4A^2)^{-1} \pmod q$; then $f(c,k) \equiv A((k-r)^2 - d)$ and $\left(\frac{A}{q}\right) = 1$, so $G = e(\ell r/q)H$ with $H = \sum_x \left(\frac{x^2-d}{q}\right) e(\ell x/q)$. If $d \equiv 0$ then $H = \sum_{x \neq 0} e(\ell x/q) = -1$. If $d \not\equiv 0$, use $\#\{y : y^2 \equiv x^2 - d\} = 1 + \left(\frac{x^2-d}{q}\right)$ and $\sum_x e(\ell x/q) = 0$ to get $H = \sum_{x^2 - y^2 \equiv d} e(\ell x/q)$. With $v = x + y$, $x - y = d\bar{v}$, this is $\sum_{v \neq 0} e(\bar{2}\ell(v + d\bar{v})/q) = S(\bar{2}\ell, \bar{2}\ell d; q)$, and Weil's bound applies since $q \nmid \ell d$.

(c) For $q \nmid \ell$ apply (b) to each class c. For $q \mid \ell$, by (a) one class contributes 0, at most one contributes $q-1$ and the others 1 each: total $\le 2q - 3$.

(d) The Jacobi symbol is $\left(\frac{\cdot}{q}\right)\left(\frac{\cdot}{q'}\right)$, $1/(qq') \equiv \bar{q}'/q + \bar{q}/q' \pmod 1$, and $f(c,k) \bmod q$ depends only on $k \bmod q$; the sum factors by the Chinese remainder theorem. Since $G_c^{(q)}$ depends only on $c \bmod q$, the sum over $c \bmod Q$ is the product of two sums as in (c). □

3.4. **A first step.** We first show how the square sieve beats the divisor bound using only Lemma 3.4(a).

Cited theorem 3.5 (Vaaler; [9, Thm. A.6], [19])**.** For every integer $H \ge 1$ there are complex numbers a_h $(1 \le |h| \le H)$ with $|a_h| \le 1/(2\pi|h|)$ such that for all real x

$$\Big|\psi(x) - \sum_{1 \le |h| \le H} a_h e(hx)\Big| \le \frac{1}{2H+2} \sum_{|h| \le H} \Big(1 - \frac{|h|}{H+1}\Big) e(hx).$$

Summing over $x = g(j)$, $j \in I$ $(|I| = N)$, and using that the right side is nonnegative, one obtains

$$\Big|\sum_{j \in I} \psi(g(j))\Big| \le \frac{N}{2H+2} + \sum_{h=1}^{H} \Big(\frac{1}{\pi h} + \frac{1}{H+1}\Big)\Big|\sum_{j \in I} e(hg(j))\Big|. \tag{6}$$

Cited theorem 3.6 (Third-derivative test; [17, Thm. 5.11])**.** Let g be real with continuous derivatives up to order three on $[a,b]$, $b - a \ge 1$, and $\lambda \le |g'''(x)| \le h\lambda$ there (with g''' of constant sign). Then $\sum_{a < j \le b} e(g(j)) \ll h^{1/2}(b-a)\lambda^{1/6} + (b-a)^{1/2}\lambda^{-1/6}$, with an absolute implied constant.

Cited theorem 3.7 (Shiu; [16, Thm. 1])**.** Fix $0 < \alpha, \beta < \frac{1}{2}$. For $1 \le k < y^{1-\alpha}$, $x^\beta < y \le x$ and $\gcd(a,k) = 1$, $\sum_{x-y < m \le x,\ m \equiv a\,(k)} \tau(m) \ll \frac{y}{\varphi(k)} \frac{1}{\log x} \exp\Big(\sum_{\ell \le x,\ \ell \mid k} \frac{2}{\ell}\Big)$.

(Shiu's theorem is stated for a class of multiplicative functions containing τ; we only need τ.)

Theorem 3.8. *For $n \ge 2$, $R(n) \ll n^{1/3 - 1/225} (\log n)^2$.*

Proof. Let $T = n^{1/3}$, $\eta = 1/75$, and assume n is large.

Small coordinates. Triples with $u < T^{1-\eta}$ number at most $\sum_{n < m \le n + T^{1-\eta}} \tau(m) \ll T^{1-\eta} \log n$ (Shiu with $k = 1$), because B is a divisor of $n + u$ and determines C; likewise for $B < T^{1-\eta}$. Triples with $C < T^{1-\eta}$ number at most $\sum_{C < T^{1-\eta}} \tau(4Cn+1)$, which is a sum of $\tau(m)$ over $m \equiv 1 \pmod{4n}$, $m \le x := 4nT^{1-\eta} + 1$. Shiu's theorem with $k = 4n$ ($k < x^{1-\alpha}$ for small α) and

$\exp(\sum_{\ell\le x,\ell\nmid k}2/\ell)\ll(\log x)^2(\varphi(k)/k)^2$ bounds this by $\ll T^{1-\eta}\log n$. Since $T^{1-\eta}=n^{1/3-1/225}$, it remains to bound the number $\mathcal{S}$ of triples with $\min(u,B,C)\ge T^{1-\eta}$.

Ranges. For such a triple let $M=\max(u,B,C)$. Then $4uBC\ge 4T^{2-2\eta}M$ and $4uBC=n+u+B\le n+2M$, so $M\le T^{1+2\eta}/2$. Hence $C\in\mathcal{C}:=[T^{1-\eta},T^{1+2\eta}/2]$ and $s=u+B\in(S_1,S_2]$ with $S_1=2T^{1-\eta}-1$, $S_2=T^{1+2\eta}$. Let $\mathcal{A}:=\{(C,k)\in\mathbb{Z}^2: C\in\mathcal{C},\ 4Ck-n\in(S_1,S_2]\}$. By Lemma 3.1,

$$\mathcal{S}\le 2\#\{(C,k)\in\mathcal{A}: f(C,k)\text{ a nonzero square}\}+\tau(n).$$

Sieve. Let $P=T^{1/25}$, $\mathcal{P}$ the primes in $(P,2P]$ and $P_0=|\mathcal{P}|\gg P/\log P$. Apply Lemma 3.2 to $w(a)=\#\{(C,k)\in\mathcal{A}: f(C,k)=a\}$ $(a\ne 0)$; the condition holds since $f\le S_2^2\le T^{2+4\eta}\le 2^{P_0}$. As $\left(\frac{0}{Q}\right)=0$,

$$\#\{(C,k)\in\mathcal{A}: f\text{ nonzero square}\}\le\frac{4}{P_0}|\mathcal{A}|+4\max_{q\ne q'\in\mathcal{P}}|S_{qq'}|,\qquad S_Q:=\sum_{(C,k)\in\mathcal{A}}\left(\frac{f(C,k)}{Q}\right).\tag{7}$$

For $C\in\mathcal{C}$ let $J_C=((n+S_1)/(4C),(n+S_2)/(4C)]$, of length $|J_C|=(S_2-S_1)/(4C)$; then $|\mathcal{A}|\le\sum_{C\in\mathcal{C}}(|J_C|+1)\ll T^{1+2\eta}\log T$.

Main term and discrepancy. Fix $Q=qq'$. For residues $c_0,k_0 \bmod Q$, the number of $k\in J_C$ with $k\equiv k_0$ is $|J_C|/Q-\psi(\beta_C)+\psi(\alpha_C)$, where $\alpha_C=((n+S_1)/(4C)-k_0)/Q$ and $\beta_C=((n+S_2)/(4C)-k_0)/Q$. As $f(C,k) \bmod Q$ depends only on $(C \bmod Q, k \bmod Q)$,

$$S_Q=\sum_{C\in\mathcal{C}}\frac{|J_C|}{Q}K(C)+\sum_{c_0,k_0 \bmod Q}\left(\frac{f(c_0,k_0)}{Q}\right)E(c_0,k_0),\quad E(c_0,k_0):=\sum_{\substack{C\in\mathcal{C}\\ C\equiv c_0\,(Q)}}(\psi(\alpha_C)-\psi(\beta_C)).$$

By Lemma 3.4(a,d), $|K(C)|\le(1+q\mathbf{1}_{D_q}(C))(1+q'\mathbf{1}_{D_{q'}}(C))$. Since $|J_C|$ decreases in C, a residue class modulo m satisfies $\sum_{C\equiv c\,(m)}|J_C|\le m^{-1}\Sigma_J+J_{\max}$, where $\Sigma_J=\sum_{C\in\mathcal{C}}|J_C|\ll T^{1+2\eta}\log T$ and $J_{\max}\le T^{3\eta}$. Expanding the product bound for $|K|$, the first sum is at most $Q^{-1}(4\Sigma_J+3QJ_{\max})\ll T^{1+2\eta}\log T/Q+T^{3\eta}$.

The discrepancy. We claim that for $Q\le T^{1/2}$

$$\sum_{c_0,k_0 \bmod Q}|E(c_0,k_0)|\ll(T^{6/7+2\eta}Q^{9/7}+T^{2/3+7\eta/3}Q^{7/6})\log T.\tag{8}$$

Split $\mathcal{C}$ into $O(\log T)$ dyadic pieces $(W,2W]$ with $T^{1-\eta}/2\le W\le T^{1+2\eta}/2$; note $Q\le T^{1/2}\le W$. Fix a piece, a class (c_0,k_0) and an endpoint $S\in\{S_1,S_2\}$. Writing $C=c_0+Qj$, j runs over an interval I of $\le 2W/Q$ integers, and the relevant ψ-values are $\psi(g(j))$ with $g(j)=(n+S)/(4Q(c_0+Qj))-k_0/Q$. For $h\ge 1$, $|(hg)'''(j)|=\frac{3}{2}h(n+S)Q^2C^{-4}\in[\lambda_h,16\lambda_h]$ with $\lambda_h=\frac{3}{32}h(n+S)Q^2W^{-4}\asymp hT^3Q^2W^{-4}$. By (6) and Cited Theorem 3.6 (an interval of length <1 contains at most one integer),

$$\Big|\sum_{j\in I}\psi(g(j))\Big|\ll\frac{W}{QH}+\sum_{h\le H}\frac{1}{h}\Big(\frac{W}{Q}\lambda_h^{1/6}+\Big(\frac{W}{Q}\Big)^{1/2}\lambda_h^{-1/6}+1\Big).$$

Here

$$\frac{W}{Q}\lambda_h^{1/6}\ll h^{\frac16}T^{\frac12}W^{\frac13}Q^{-\frac23}\le h^{\frac16}T^{\frac56+\frac{2\eta}{3}}Q^{-\frac23},\qquad\Big(\frac{W}{Q}\Big)^{\frac12}\lambda_h^{-\frac16}\ll W^{\frac76}T^{-\frac12}Q^{-\frac56}\le T^{\frac23+\frac{7\eta}{3}}Q^{-\frac56}.$$

Summing over h and choosing $H=\lceil T^{1/7}Q^{-2/7}\rceil$ gives $\ll T^{6/7+2\eta}Q^{-5/7}+T^{2/3+7\eta/3}Q^{-5/6}+\log T$ for each piece, class and endpoint. Multiplying by Q^2 classes, two endpoints and $O(\log T)$ pieces gives (8) (the $Q^2\log^2 T$ from the term $\log T$ is dominated).

Assembly. $Q \le 4P^2 \le T^{1/2}$. By the above, $\max_Q |S_Q| \ll T^{1+2\eta}\log T/P^2 + T^{3\eta} + (T^{6/7+2\eta}P^{18/7} + T^{2/3+7\eta/3}P^{7/3})\log T$, and so

$$\mathcal{S} \ll \frac{\log P}{P} T^{1+2\eta}\log T + T^{6/7+2\eta}P^{18/7}\log T + T^{2/3+7\eta/3}P^{7/3}\log T + \tau(n).$$

With $P = T^{1/25}$ the first two terms are $\ll T^{24/25+2\eta}(\log T)^2 = T^{74/75}(\log T)^2$, since $\frac{6}{7}+\frac{18}{175} = \frac{24}{25}$; the third is $\le T^{0.88}$. As $T^{74/75} = n^{1/3-1/225}$, the theorem follows. □

Remark 3.9. The same argument restricted to the cube block $U = V \asymp (n/4)^{1/3}$ (where $\eta = 0$ in all size bounds) gives $N(U,U,W) \ll n^{8/25}(\log n)^2$. The loss in (8) comes from taking absolute values over all Q^2 residue classes. The next subsections avoid this by opening the character in the k-variable.

3.5. **Poisson summation in k.** Fix $\Phi \in C_c^\infty(\mathbb{R})$ with $0 \le \Phi \le 1$, $\Phi = 1$ on $[1,4]$ and $\operatorname{supp}\Phi \subset [\frac12, 5]$. Let $\hat\Phi(\xi) = \int \Phi(t)e(-\xi t)\,dt$ and $\Psi(x) := \sup_{|\xi|\ge|x|}(|\hat\Phi(\xi)| + |\xi\hat\Phi'(\xi)|)$; then Ψ is non-increasing in $|x|$ and $\Psi(x) \ll_A (1+|x|)^{-A}$ for every A. Write $\chi_Q = \left(\frac{\cdot}{Q}\right)$ for $Q = qq'$, and $\chi_1 \equiv 1$, $G^{(1)} \equiv 1$.

Lemma 3.10. *For integers $C \ge 1$, $Q \in \{1\}\cup\{qq'\}$ and real $V > 0$,*

$$\sum_{k\in\mathbb{Z}} \Phi\Big(\frac{4Ck-n}{V}\Big)\chi_Q(f(C,k)) = \sum_{\ell\in\mathbb{Z}} a_\ell(C)\, e\Big(-\frac{\ell n}{4CQ}\Big) G_C^{(Q)}(\ell), \qquad a_\ell(C) := \frac{V}{4CQ}\hat\Phi\Big(\frac{\ell V}{4CQ}\Big).$$

Proof. Write $k = k_0 + Qj$ with $k_0 \bmod Q$; $\chi_Q(f(C,k))$ depends only on k_0. Poisson summation in j applied to $x \mapsto \Phi((4CQx + 4Ck_0 - n)/V)$, with the substitution $t = (4CQx + 4Ck_0 - n)/V$, gives $\sum_\ell \frac{V}{4CQ}\hat\Phi(\frac{\ell V}{4CQ})e(-\frac{\ell n}{4CQ})e(\frac{\ell k_0}{Q})$. Multiply by $\chi_Q(f(C,k_0))$ and sum over k_0. The ℓ-series converges absolutely. □

Proposition 3.11. *Let $V, W \ge 1$, $Q \in \{1\}\cup\{qq'\}$, and*

$$S_Q := \sum_{C\in(W,2W]}\sum_{k\in\mathbb{Z}} \Phi\Big(\frac{4Ck-n}{V}\Big)\chi_Q(f(C,k)).$$

Then, with an implied constant depending only on Φ,

$$|S_Q| \ll \frac{V}{Q} + \frac{V}{W} + Q\, n^{1/6}W^{1/2}V^{-1/6} + Q^{1/2}V^{1/6}W\, n^{-1/6} + Q^{3/2}. \tag{9}$$

Proof. By Lemma 3.10, $S_Q = \sum_\ell \sum_{C\in(W,2W]} a_\ell(C)e(\varphi_\ell(C))G_C(\ell)$ with $\varphi_\ell(C) = -\ell n/(4CQ)$ and $G = G^{(Q)}$.

(i) $\ell = 0$. $G_C(0) = K(C)$ (or 1 if $Q = 1$), and $|a_0(C)| \le V\hat\Phi(0)/(4QW)$. By Lemma 3.4(a,d), $\sum_{C\in(W,2W]}|K(C)| \le W + q(\frac{W}{q}+1) + q'(\frac{W}{q'}+1) + Q(\frac{W}{Q}+1) \le 4W + 3Q$, so the $\ell = 0$ term is $\ll V/Q + V/W$ (and $\ll V$ if $Q = 1$).

(ii) $\ell \ne 0$. Grouping C by $c_0 \equiv C \pmod Q$, the ℓ-th term is at most $\sum_{c_0 \bmod Q}|G_{c_0}(\ell)| \cdot |\sum_{C\equiv c_0} a_\ell(C)e(\varphi_\ell(C))|$. For $C \in (W,2W]$ we have $|\xi| \ge |\ell|V/(8QW)$ where $\xi = \ell V/(4CQ)$, hence $|a_\ell(C)| \le \frac{V}{4QW}\Psi(\frac{\ell V}{8QW})$; and $\frac{d}{dC}a_\ell(C) = -\frac{V}{4QC^2}(\hat\Phi(\xi)+\xi\hat\Phi'(\xi))$, so $\int_W^{2W}|a_\ell'| \le \frac{V}{8QW}\Psi(\frac{\ell V}{8QW})$. Partial summation gives

$$\Big|\sum_{C\equiv c_0} a_\ell(C)e(\varphi_\ell(C))\Big| \le \frac{3V}{8QW}\Psi\Big(\frac{\ell V}{8QW}\Big)E_Q(\ell), \quad E_Q(\ell) := \max_{c_0}\max_J\Big|\sum_{\substack{C\in J\\ C\equiv c_0\,(Q)}} e(\varphi_\ell(C))\Big|,$$

J running over subintervals of $(W, 2W]$. With Lemma 3.4(d),

$$|S_Q - (\ell = 0 \text{ term})| \le \frac{3V}{2QW} Q^{3/2} \sum_{\ell \neq 0} \Psi\Big(\frac{\ell V}{8QW}\Big) E_Q(\ell).$$

(iii) Third-derivative test. Write $C = c_0 + Qj$; j runs over an interval of length $\le W/Q$, and $g(j) = \varphi_\ell(c_0 + Qj)$ has $g'''(j) = \frac{3}{2}\ell n Q^2 C^{-4}$, so $\lambda_3 \le |g'''| \le 16\lambda_3$ with $\lambda_3 = \frac{3}{32}|\ell| n Q^2 W^{-4}$. Cited Theorem 3.6 (and the trivial bound for intervals of length < 1) gives

$$E_Q(\ell) \ll |\ell|^{1/6} n^{1/6} W^{1/3} Q^{-2/3} + |\ell|^{-1/6} n^{-1/6} W^{7/6} Q^{-5/6} + 1.$$

(iv) The ℓ-sum. For $\sigma \in \{\frac{1}{6}, 0, -\frac{1}{6}\}$ and $M > 0$ we have $\sum_{\ell \neq 0} \Psi(\ell/M)|\ell|^\sigma \ll M^{1+\sigma}$ (for $M \ge 1$ split at $|\ell| = M$ and use $\Psi(x) \ll |x|^{-2}$; for $M < 1$ use $\Psi(\ell/M) \ll M^2 \ell^{-2} \le M^{1+\sigma}\ell^{-2}$). With $M = 8QW/V$:

$$\frac{V}{QW} Q^{3/2}\Big[\Big(\frac{QW}{V}\Big)^{7/6} n^{1/6} W^{1/3} Q^{-2/3} + \Big(\frac{QW}{V}\Big)^{5/6} n^{-1/6} W^{7/6} Q^{-5/6} + \frac{QW}{V}\Big]$$
$$= Q n^{1/6} W^{1/2} V^{-1/6} + Q^{1/2} V^{1/6} W n^{-1/6} + Q^{3/2}. \quad \square$$

3.6. **Proof of Theorem A.**

Theorem 3.12. *For every $\varepsilon > 0$ and $n \ge 2$, $R(n) \ll_\varepsilon n^{12/37+\varepsilon}$. The cube block satisfies $N(U,U,U) \ll_\varepsilon n^{17/54+\varepsilon}$.*

Proof. It suffices to bound $N(U,V,W)$ for each of the $O((\log n)^3)$ nonempty blocks; by symmetry (Lemma 2.1(ii)) assume $U \le V$. Put $\alpha = \log U/\log n$, $\beta = \log V/\log n$, $\gamma = \log W/\log n$; then $|\alpha+\beta+\gamma-1| \le 5/\log n$. Let $\theta^* = 12/37$.

Case 1: $\min(\alpha,\gamma) \le \theta^*$. Lemma 2.4 gives $N \ll_\varepsilon n^{\theta^*+\varepsilon}$.

Case 2: $\alpha, \gamma > \theta^*$. Then $\beta \ge \alpha > \theta^*$. *Smoothed detection.* Put $\omega(C,k) = \mathbf{1}_{(W,2W]}(C)\Phi((4Ck-n)/V)$ and $w(a) = \sum_{f(C,k)=a} \omega(C,k)$ for $a \neq 0$. A block triple with $u \neq B$ has $s = u + B \in (U+V, 2U+2V] \subset (V, 4V]$, so its image (C,k) under Lemma 3.1 has $\omega(C,k) = 1$ and $f(C,k) \ge 1$. Hence $N \le 2\sum_{m \ge 1} w(m^2) + \tau(n)$. If $\omega(C,k) \neq 0$ then $4Ck - n \in [V/2, 5V]$, so $k \ge 1$ and $f(C,k) < (4Ck-n)^2 \le 25V^2$.

Parameters. Let $\rho^* := \frac{1}{3}(\frac{7}{6}\beta - \frac{1}{6} - \frac{\gamma}{2})$, $P = n^{\rho^*}$, and $\mathcal{P}$ the primes in $(P, 2P]$; so $Q = qq' \in (P^2, 4P^2]$. We check below that $\rho^* \ge 4/333$, so $P_0 = |\mathcal{P}| \gg n^{\rho^*}/\log n$ and $2^{P_0} \ge 25V^2$ for $n \ge n_0$. Lemma 3.2 and Proposition 3.11 (with $Q = 1$ for $\sum_a w(a) \le S_1$, and $\sum_a w(a)\chi_Q(a) = S_Q$ as $\chi_Q(0) = 0$) give

$$N \ll \frac{\log n}{P}\Big[V + \frac{V}{W} + n^{1/6} W^{1/2} V^{-1/6} + V^{1/6} W n^{-1/6} + 1\Big]$$
$$+ \frac{V}{P^2} + \frac{V}{W} + P^2 n^{1/6} W^{1/2} V^{-1/6} + P V^{1/6} W n^{-1/6} + P^3 + \tau(n).$$

Exponents. Up to $O(1/\log n)$ and the factor $\log n$, the terms have exponents (base n)

$$\beta - \rho^*,\ \beta - \gamma - \rho^*,\ d - \rho^*,\ e - \rho^*,\ -\rho^*,\ \beta - 2\rho^*,\ \beta - \gamma,\ 2\rho^* + d,\ \rho^* + e,\ 3\rho^*,$$

where $d = \frac{1}{6} + \frac{\gamma}{2} - \frac{\beta}{6}$ and $e = \frac{\beta}{6} + \gamma - \frac{1}{6}$, and $\tau(n) = n^{o(1)}$. The choice of ρ^* balances $\beta - \rho^* = 2\rho^* + d$, and with $\beta = 1 - \alpha - \gamma$,

$$\beta - \rho^* = \frac{11\beta + 1 + 3\gamma}{18} = \frac{12 - 11\alpha - 8\gamma}{18} < \frac{12 - 19\theta^*}{18} = \theta^*.$$

After substituting $\beta = 1 - \alpha - \gamma$, every exponent in the list is an affine function of (α, γ), and the region $\{\alpha \ge \theta^*, \gamma \ge \theta^*, \beta \ge \alpha\}$ is the triangle with vertices $(\alpha,\beta,\gamma) = (\frac{12}{37}, \frac{13}{37}, \frac{12}{37})$, $(\frac{12}{37}, \frac{12}{37}, \frac{13}{37})$, $(\frac{25}{74}, \frac{25}{74}, \frac{12}{37})$. At these vertices the maximum of the listed exponents is $\frac{12}{37}$, $\frac{104}{333}$, $\frac{421}{1332}$

respectively (exact rational arithmetic; the maximum is attained by $\beta-\rho^*$ and $2\rho^*+d$), and ρ^* equals $\frac{1}{37}$, $\frac{4}{333}$, $\frac{29}{1332}$; in particular $\rho^*\ge\frac{4}{333}$ on the triangle. The $O(1/\log n)$ discrepancy between $\alpha+\beta+\gamma$ and 1 changes each term by a bounded factor. Hence $N\ll_\varepsilon n^{12/37+\varepsilon}$ in Case 2. At the cube $\alpha=\beta=\gamma=\frac{1}{3}$ we get $\rho^*=\frac{1}{54}$ and $N\ll_\varepsilon n^{17/54+\varepsilon}$. □

Remark 3.13. The worst block is not the cube but $(\alpha,\beta,\gamma)=(\frac{12}{37},\frac{13}{37},\frac{12}{37})$: the sieve's main term is $|\mathcal{A}|/P\asymp V/P$ with $V=\max(U,V)$, while the trivial bound is $\min(U,W)$.

3.7. **Exponent pairs.** We use the definition of exponent pairs of Graham and Kolesnik [9, Ch. 3]: (κ,λ) is an exponent pair if for every $\sigma>0$ there are P and $c<\frac{1}{2}$ such that $\sum_{j\in I}e(g(j))\ll(y/N^\sigma)^\kappa N^\lambda$ whenever $y\ge N^\sigma$, $I\subseteq[N,2N]$ is an interval, and g has P continuous derivatives with $|g^{(r+1)}(x)-\frac{d^r}{dx^r}(yx^{-\sigma})|\le c|\frac{d^r}{dx^r}(yx^{-\sigma})|$ on I for $0\le r\le P-1$. (We take this formulation from [18, §1.1], which restates [9, Ch. 3].) The pair $(\frac{1}{6},\frac{2}{3})=AB(0,1)=A(\frac{1}{2},\frac{1}{2})$ is an exponent pair [9, Ch. 3], [18, §1.2].

Theorem 3.14. *Let $(\kappa,\kappa+\frac{1}{2})$ be an exponent pair with $0\le\kappa<\frac{1}{4}$. Then $R(n)\ll_\varepsilon n^{4/(13-4\kappa)+\varepsilon}$.*

Proof. Put $\theta^*=4/(13-4\kappa)\in(\frac{1}{4},\frac{1}{3})$ and argue as in the proof of Theorem 3.12, replacing step (iii) of Proposition 3.11 by the exponent pair. In Case 2 we have $\gamma\le1-2\theta^*+o(1)<\frac{1}{2}$, so $|\ell|n\ge4W^2\cdot n^c$ for every $\ell\ne0$ and some $c>0$. For $\ell>0$ the sum defining $E_Q(\ell)$ is $\sum_j e(g(j))$ with $g(x)=-Y/(x+a)$, $Y=\ell n/(4Q^2)$, $a=c_0/Q\in[0,1)$ (taking $1\le c_0\le Q$), and $j+a\in(W/Q,2W/Q]$. After removing at most one boundary term, j lies in an interval $I\subseteq[N,2N]$, $N=\lceil W/Q\rceil$, and $g'(x)=Y(x+a)^{-2}$ satisfies the derivative conditions with $\sigma=2$, $y=Y$, since $(x/(x+a))^{2+r}=1+O_r(1/N)$; this needs $N\ge N_0$, and for $N<N_0$ we use the trivial bound. Also $y\ge N^2$ because $\ell n\ge4W^2$. The case $\ell<0$ follows by complex conjugation. Hence $E_Q(\ell)\ll(|\ell|n/W^2)^\kappa(W/Q)^{\kappa+1/2}+1$, and step (iv) gives the contribution $\frac{V}{QW}Q^{3/2}(n/W^2)^\kappa(W/Q)^{\kappa+1/2}M^{1+\kappa}\asymp Q\,n^\kappa W^{1/2}V^{-\kappa}$ in place of the two third-derivative terms of (9). Choose $\rho^*=\frac{1}{3}((1+\kappa)\beta-\kappa-\frac{\gamma}{2})$, which balances $\beta-\rho^*$ against $2\rho^*+\kappa+\frac{\gamma}{2}-\kappa\beta$; then $\beta-\rho^*=\frac{1}{6}(4-(4-2\kappa)\alpha-(3-2\kappa)\gamma)$, which is $<\theta^*$ for $\alpha,\gamma>\theta^*$ exactly when $\theta^*\ge4/(13-4\kappa)$. The remaining terms are $\beta-\gamma-\rho^*$, $\kappa+\frac{\gamma}{2}-\kappa\beta-\rho^*$, $-\rho^*$, $\beta-2\rho^*$, $\beta-\gamma$, $3\rho^*$. As before they are affine on the triangle $\{\alpha,\gamma\ge\theta^*,\ \beta\ge\alpha\}$, whose vertices are $(\theta^*,1-2\theta^*,\theta^*)$, $(\theta^*,\theta^*,1-2\theta^*)$, $(\frac{1-\theta^*}{2},\frac{1-\theta^*}{2},\theta^*)$. At the vertices one checks (exact arithmetic as functions of κ; the deficits θ^*-(term) are nonnegative for $0\le\kappa<\frac{1}{4}$) that all terms are $\le\theta^*$ and $0<2\rho^*<\gamma$. □

Remark 3.15. For $\kappa=\frac{1}{6}$ this recovers $\frac{12}{37}$. Bourgain [2, Thm. 6] states that $(\frac{13}{84}+\varepsilon,\frac{55}{84}+\varepsilon)$ is an exponent pair, for phase functions "in the class of functions to which the exponent pair theory applies" with reference to [9, Ch. 3]; his proof of Theorem 6 is a sketch. Our phases $-Y/(x+a)$ are perturbations of the model phase $-Y/x$ of the Graham–Kolesnik class. Relative to Bourgain's theorem for these phases, Theorem 3.14 gives $R(n)\ll_\varepsilon n^{21/65+\varepsilon}$ ($21/65=0.3230\ldots$); we do not re-verify Bourgain's argument. If $(\varepsilon,\frac{1}{2}+\varepsilon)$ is an exponent pair for every $\varepsilon>0$ (the exponent pair conjecture), we get $R(n)\ll_\varepsilon n^{4/13+\varepsilon}$.

Remark 3.16 (where the method stops)**.** The bound beats the trivial exponent $\min(\alpha,\gamma)$ only near the cube; on the diagonal $\alpha=\beta$ the gain region of the bound in the proof of Theorem 3.12 is $\beta\in(\frac{4}{13},\frac{14}{41})$ (a computation with its exponents). At the "balanced" block $u,B\asymp n^{1/4}$, $C\asymp n^{1/2}$ the strip has $O(1)$ points per value of C ($|J_C|\asymp V/W\asymp n^{-1/4}$), and summing the dual frequencies ℓ before C undoes the Poisson step. Heuristically: for most classes c_0 one has $|G_{c_0}(\ell)|\asymp\sqrt{Q}$ on average over ℓ, a bound for the C-sums that is uniform in many ℓ cannot beat $\sqrt{W/Q}$ (by Fejér's kernel), and inserting these sizes into step (ii) of Proposition 3.11 gives

a dual contribution $\gg QW^{1/2}$. The block bound is then $\gg \max(V/P, P^2W^{1/2}) \geq V^{2/3}W^{1/6}$, which at the balanced block equals the trivial bound V. So with the triangle inequality over ℓ and c_0, even square-root cancellation in every C-sum would not help there; one would need cancellation in the bilinear sum $\sum_{\ell,C} G_C(\ell)e(\ell n/(4CQ))$. We do not formalise or pursue this.

4. The genus main term

Throughout this section $p \equiv 1 \pmod 4$ is prime, $M_{\mathrm{gen}}(p)$ is defined by (4), and ℓ always denotes a prime. The interpretation of M_{gen} is as follows. For $(u,B) \in \mathcal{P}(p)$ both p and a are units modulo q (Lemma 2.2), and $\mathbf{1}[p \equiv a\ (q)] = \varphi(q)^{-1}\sum_{\chi \bmod q} \chi(p)\bar{\chi}(a)$. The real characters modulo q are the characters of $(\mathbb{Z}/q\mathbb{Z})^\times/((\mathbb{Z}/q\mathbb{Z})^\times)^2$, so their part of the sum is $\varphi(q)^{-1}\sum_{\chi^2=1}\chi(pa) = \frac{s(q)}{\varphi(q)}\mathbf{1}[pa \in ((\mathbb{Z}/q\mathbb{Z})^\times)^2]$. Summing over $\mathcal{P}(p)$ gives $M_{\mathrm{gen}}(p)$, and $R(p) - M_{\mathrm{gen}}(p)$ is the contribution of characters of order ≥ 3.

Remark 4.1. For every pair with $\gcd(u,B) = 1$ and $u+B$ odd, the Kronecker character χ_{-4uB} (a character modulo $q = 4uB$) satisfies $\chi_{-4uB}(a) = -1$: indeed $\chi_{-4uB}(-1) = -1$, and since $B \equiv -u \pmod{u+B}$, $\chi_{-4uB}(u+B) = \left(\frac{-uB}{u+B}\right) = \left(\frac{u^2}{u+B}\right) = 1$. So a is never a square modulo q. Consequently, if p is replaced by an odd square m^2 in (4) (terms with $\gcd(m,q) > 1$ being 0), every term vanishes: m^2a is a square unit only if a is. Only the single character χ_{-4uB}, not the full real-character expansion, is needed for this vanishing.

4.1. **A genus-exact subfamily.** For a squarefree $u \equiv 1 \pmod 4$ with $p \nmid u$ put $D = D_u := -4up$, a fundamental discriminant ($-up \equiv 3 \pmod 4$ and up is squarefree). Let $w_0 := -(p+u)/2$, an odd integer, and

$$\mathfrak{E}_u := \{\eta_{d,j} := \left(\frac{\cdot}{d}\right)\chi_{-4}^j :\ d \mid u,\ j \in \{0,1\}\}, \qquad \varepsilon_{d,j} := \left(\frac{-2p}{d}\right)\chi_{-4}(w_0)^j \in \{\pm 1\},$$

where $\left(\frac{\cdot}{d}\right)$ is the Jacobi symbol (a primitive character of conductor d for squarefree odd d; trivial for $d = 1$). Note $\eta_{1,0} = 1$ and $\varepsilon_{1,0} = 1$. For odd w coprime to up put

$$\rho_u(w) := \frac{2^{\omega(w)}}{\varphi(w)}\,\mathbf{1}[\chi_D(\ell) = 1 \text{ for every prime } \ell \mid w].$$

Lemma 4.2. *Let $u \equiv 1 \pmod 4$ be squarefree and w odd with $\gcd(w,u) = 1$, $p \nmid uw$ and $8uw \leq p$. Then $(u, 2w)$ and $(2w, u)$ are distinct elements of $\mathcal{P}(p)$, both with $q = 8uw$ and $a = -(u+2w)$, and each contributes*

$$\frac{2^{\omega(u)}}{\varphi(u)}\,\rho_u(w)\,2^{-\omega(u)-1}\sum_{\eta\in\mathfrak{E}_u}\varepsilon_\eta\,\eta(w)$$

to $M_{\mathrm{gen}}(p)$. Hence the two together contribute $\varphi(u)^{-1}\rho_u(w)\sum_{\eta\in\mathfrak{E}_u}\varepsilon_\eta\eta(w)$.

Proof. $4u\cdot 2w - u - 2w < 8uw \leq p$, $\gcd(u,2w) = 1$ and $u+2w$ is odd. As $8 \parallel q$ and uw is odd, $(\mathbb{Z}/q\mathbb{Z})^\times \cong (\mathbb{Z}/8\mathbb{Z})^\times \times \prod_{\ell\mid uw}(\mathbb{Z}/\ell^{v_\ell(uw)}\mathbb{Z})^\times$, so $s(q) = 4\cdot 2^{\omega(uw)}$ and $\varphi(q) = 4\varphi(u)\varphi(w)$. A unit is a square modulo q iff it is $\equiv 1 \pmod 8$ and a quadratic residue modulo every prime $\ell \mid uw$. For $\ell \mid w$, $a \equiv -u$, and $\left(\frac{-up}{\ell}\right) = \chi_D(\ell)$; for $\ell \mid u$, $a \equiv -2w$; and $pa \equiv 1 \pmod 8$ iff $a \equiv p \pmod 8$ iff $2w \equiv -(p+u) \pmod 8$ iff $w \equiv w_0 \pmod 4$ (note $p+u \equiv 2 \pmod 4$). So the genus condition is: $\chi_D(\ell) = 1$ for all $\ell \mid w$; $\left(\frac{-2pw}{\ell}\right) = 1$ for all $\ell \mid u$; and $\chi_{-4}(w) = \chi_{-4}(w_0)$. Since $\gcd(w,u) = 1$ the last two conditions have indicator $\prod_{\ell\mid u}\frac{1}{2}(1 + \left(\frac{-2p}{\ell}\right)\left(\frac{w}{\ell}\right)) \cdot \frac{1}{2}(1 + \chi_{-4}(w_0)\chi_{-4}(w)) = 2^{-\omega(u)-1}\sum_{\eta\in\mathfrak{E}_u}\varepsilon_\eta\eta(w)$. □

4.2. **Analytic inputs.**

Cited theorem 4.3 (Burgess [4, 5]; [13, Thm. 12.6])**.** For every integer $r \geq 1$ and $\varepsilon > 0$ there is $C(r,\varepsilon)$ such that for every primitive Dirichlet character χ of cube-free conductor $f > 1$, all real $M \geq 0$ and $N \geq 1$,

$$\Big|\sum_{M<n\leq M+N} \chi(n)\Big| \leq C(r,\varepsilon) N^{1-1/r} f^{(r+1)/(4r^2)+\varepsilon}.$$

The constant $C(r,\varepsilon)$ is effectively computable. For $r = 1$ this is the Pólya–Vinogradov inequality. For $r \geq 2$ an explicit version is [10, Thm. 1.2] (in v2: for $f \geq \max(10^{1143}, 2^{4r-2})$, with additional factors that are $O_{r,\varepsilon}(f^{\varepsilon})$ effectively); smaller f are covered by $|\sum \chi(n)| \leq \min(N, f) \leq f^{1/r} N^{1-1/r}$.

Fix an integer $r \geq 3$ and $\varepsilon \in (0, \frac{1}{100})$, and put

$$\kappa := \frac{1}{r}, \qquad \theta := \frac{r+1}{4r^2} + \varepsilon, \qquad \varpi := r\theta = \frac{r+1}{4r} + r\varepsilon.$$

Implied constants in this section depend only on r, ε and on fixed exponents introduced below; all of them are effective.

Lemma 4.4. *Let $\chi = \chi^* \cdot \mathbf{1}[\gcd(\cdot, e) = 1]$ with χ^* primitive of cube-free conductor $f > 1$ and $e \geq 1$. Put $K := (1+r)\tau(e)C(r,\varepsilon)f^{\theta}$. Then for $t \geq 1$*

$$\Big|\sum_{n>t} \frac{\chi(n)}{n}\Big| \leq K t^{-\kappa}, \qquad \Big|\sum_{n>t} \frac{\chi(n)\log n}{n}\Big| \leq (r+1) K t^{-\kappa}(1+\log t).$$

Proof. Replacing e by its radical, $S(v) := \sum_{n\leq v} \chi(n) = \sum_{g\mid e} \mu(g)\chi^*(g) \sum_{m\leq v/g} \chi^*(m)$, so $|S(v)| \leq \tau(e) C(r,\varepsilon) v^{1-\kappa} f^{\theta}$. By partial summation $\sum_{n>t} \chi(n)/n = -S(t)/t + \int_t^{\infty} S(v) v^{-2} dv$, which is at most $\tau(e) C f^{\theta}(t^{-\kappa} + r t^{-\kappa})$. Similarly, with $d(\log v/v) = (1 - \log v) v^{-2} dv$ and $\int_t^{\infty} v^{-1-\kappa}(1+\log v) dv = r t^{-\kappa}(1 + r + \log t)$, the second sum is at most $\tau(e) C f^{\theta} t^{-\kappa}(1+r)(r + \log t)$. □

For arithmetic functions F, G and $y > 0$ write $\Phi_{F,G}(y) := \sum_{ab\leq y} F(a)G(b)/(ab)$ (so $\Phi(y) = 0$ for $y < 1$), $L(1,F) = \sum F(a)/a$ and $L'(1,G) = -\sum_b G(b)\log b/b$ when these converge.

Lemma 4.5 (hyperbola method)**.** (a) *Let F, G be as in Lemma 4.4 with constants $K_F, K_G \geq 1$. If $y^{\kappa} \geq K_F K_G$ then $|\Phi_{F,G}(y) - L(1,F)L(1,G)| \leq (2r+3)(K_F K_G y^{-\kappa})^{1/2}$.*

(b) *Let $F \equiv 1$ and G as in Lemma 4.4 with constant K_G. For $1 \leq z \leq y$,*

$$\big|\Phi_{1,G}(y) - L(1,G)(\log y + \gamma_E) - L'(1,G)\big| \leq 5r^2 K_G (z/y)^{\kappa}(1+\log y) + 2/z.$$

Proof. Let $\mathfrak{F}(t) = \sum_{a\leq t} F(a)/a$, $\mathfrak{G}(t) = \sum_{b\leq t} G(b)/b$. For $1 \leq z \leq y$, Dirichlet's hyperbola identity is $\Phi(y) = \sum_{a\leq z} \frac{F(a)}{a} \mathfrak{G}(\frac{y}{a}) + \sum_{b\leq y/z} \frac{G(b)}{b} \mathfrak{F}(\frac{y}{b}) - \mathfrak{F}(z)\mathfrak{G}(\frac{y}{z})$.

(a) Write $\mathfrak{F} = L_F - r_F$, $\mathfrak{G} = L_G - r_G$, with $|r_F(t)| \leq K_F t^{-\kappa}$, $|r_G(t)| \leq K_G t^{-\kappa}$ for $t \geq 1$. Substituting, all products of L's combine to $L_F L_G$ and

$$\Phi(y) - L_F L_G = -\sum_{a\leq z} \frac{F(a)}{a} r_G\Big(\frac{y}{a}\Big) - \sum_{b\leq y/z} \frac{G(b)}{b} r_F\Big(\frac{y}{b}\Big) - r_F(z) r_G\Big(\frac{y}{z}\Big).$$

As $\sum_{a\leq z} a^{\kappa-1} \leq 1 + r z^{\kappa}$, the three terms are at most $K_G y^{-\kappa}(1 + r z^{\kappa})$, $K_F y^{-\kappa}(1 + r(y/z)^{\kappa})$ and $K_F K_G y^{-\kappa}$. Take $z = y^{1/2}(K_F/K_G)^{r/2}$; the hypothesis and $K_F, K_G \geq 1$ give $1 \leq z \leq y$. Then $r K_G y^{-\kappa} z^{\kappa} = r K_F z^{-\kappa} = r(K_F K_G y^{-\kappa})^{1/2}$, and the remaining terms total $\leq 3 K_F K_G y^{-\kappa} \leq 3(K_F K_G y^{-\kappa})^{1/2}$.

(b) Now $\mathfrak{F}(t)=\log t+\gamma_E+r_1(t)$ with $|r_1(t)|\le 1/t$ for $t\ge 1$. Write $r_G'(t)=\sum_{b>t}G(b)\log b/b$. Substituting and collecting terms,

$$\Phi(y)=L_G(\log y+\gamma_E)+L'(1,G)-\sum_{a\le z}\frac{1}{a}r_G\Big(\frac{y}{a}\Big)-\log\frac{y}{z}\,r_G\Big(\frac{y}{z}\Big)+r_G'\Big(\frac{y}{z}\Big)$$
$$+\sum_{b\le y/z}\frac{G(b)}{b}r_1\Big(\frac{y}{b}\Big)+r_1(z)r_G\Big(\frac{y}{z}\Big).$$

(The coefficient of $\log z$ cancels between $L_G\mathfrak{F}(z)$ and $-\mathfrak{F}(z)L_G$; the b-sum gives $(\log y+\gamma_E)(L_G-r_G(y/z))+L'(1,G)+r_G'(y/z)$.) The five error terms are bounded by $K_Gy^{-\kappa}(1+rz^\kappa)$, $K_G(z/y)^\kappa\log y$, $(r+1)K_G(z/y)^\kappa(1+\log y)$, $\sum_{b\le y/z}b^{-1}(b/y)\le 1/z$ and $K_G(z/y)^\kappa/z$; their sum is at most the stated bound. □

4.3. **Windowed sums: only the principal genus character survives.** Fix exponents $0<\alpha_0<\alpha_1$ and $\beta<\gamma$, put $U_i=p^{\alpha_i}$, $Y=p^\beta$, $X=p^\gamma$, and

$$\mathcal{U}:=\{u\text{ squarefree}:\ u\equiv 1\ (4),\ U_0<u\le U_1\},\qquad \mathcal{S}_u:=\{w\text{ odd squarefree}:\ \gcd(w,up)=1\}.$$

For $u\in\mathcal{U}$ and $\eta\in\mathfrak{E}_u$ let $S_\eta(u):=\sum_{w\in\mathcal{S}_u\cap(Y,X]}\eta(w)\rho_u(w)$.

Proposition 4.6. *Assume $\gamma<1-\alpha_1$ and $\beta>\varpi(1+\alpha_1)+4\varepsilon$. There is $\delta=\delta(r,\varepsilon,\alpha_1,\beta)>0$ such that, uniformly for $u\in\mathcal{U}$ and $\eta\in\mathfrak{E}_u$,*

$$S_\eta(u)=\mathbf{1}[\eta=1]\,H_u\,L(1,\chi_{D_u})\,(\gamma-\beta)\log p+O(p^{-\delta}),\qquad H_u:=\frac{6}{\pi^2}\prod_{\ell\mid 2up}\frac{\ell}{\ell+1}.$$

Proof. (1) Factorisation. For $w\in\mathcal{S}_u$, $\sum_{b\mid w}\chi_D(b)=\prod_{\ell\mid w}(1+\chi_D(\ell))=2^{\omega(w)}\mathbf{1}[\chi_D(\ell)=1\ \forall\ell\mid w]$. So $\rho_u(w)=g(w)/w$ with $g(w):=\mathbf{1}_{\mathcal{S}_u}(w)\frac{w}{\varphi(w)}\sum_{b\mid w}\chi_D(b)$, a multiplicative function. Define h by $g=1*\chi_D*h$. Comparing local factors (with $Y_\ell=\ell^{-s}$): for $\ell\mid 2up$ we have $\chi_D(\ell)=0$ and $h(\ell)=-1$, $h(\ell^k)=0$ for $k\ge 2$; for $\chi_D(\ell)=1$ the factor of h is $(1+\frac{2\ell}{\ell-1}Y_\ell)(1-Y_\ell)^2$, so $h(\ell)=\frac{2}{\ell-1}$, $h(\ell^2)=-\frac{3\ell+1}{\ell-1}$, $h(\ell^3)=\frac{2\ell}{\ell-1}$; for $\chi_D(\ell)=-1$ it is $(1-Y_\ell)(1+Y_\ell)$, so $h(\ell^2)=-1$ and the other values vanish. Consequently, for fixed $0\le\theta'<\frac12$,

$$\sum_{c\ge1}|h(c)|c^{\theta'-1}\le 2^{\omega(2up)}\prod_{\ell\ge3}(1+2\ell^{\theta'-1}/(\ell-1)+5\ell^{2\theta'-2}+3\ell^{3\theta'-3})=p^{o(1)}.$$

At $s=1$ each factor with $\ell\nmid 2up$ equals $1-\ell^{-2}$, so $\sum_c h(c)/c=\prod_{\ell\mid 2up}(1-\ell^{-1})\prod_{\ell\nmid 2up}(1-\ell^{-2})=H_u$. Since η is completely multiplicative,

$$S_\eta(u)=\sum_{c\ge1}\frac{\eta(c)h(c)}{c}\,[\Phi_\eta(X/c)-\Phi_\eta(Y/c)],\qquad \Phi_\eta:=\Phi_{\eta,\eta\chi_D}.$$

(2) The characters. For odd n, $\chi_D(n)=\chi_{-4}(n)\left(\frac{n}{u}\right)\left(\frac{n}{p}\right)$ by quadratic reciprocity ($u\equiv p\equiv 1\pmod 4$), and $\chi_D(n)=0$ for even n. Hence for $\eta=\eta_{d,j}$

$$\eta\chi_D(n)=\chi_{-4}(n)^{j+1}\left(\frac{n}{u/d}\right)\left(\frac{n}{p}\right)\mathbf{1}[\gcd(n,2d)=1],$$

so $\eta\chi_D=\chi^*\mathbf{1}[\gcd(\cdot,2d)=1]$ with χ^* primitive of conductor $f_G=4^{1-j}(u/d)p$. Since $p\mid f_G$, *$\eta\chi_D$ is non-principal for every $\eta\in\mathfrak{E}_u$.* The character η is principal iff $(d,j)=(1,0)$, and otherwise primitive of conductor $f_F=4^jd$. In all cases $f_Ff_G=4up$, and both conductors are cube-free.

(3) $\eta\ne1$. Lemma 4.4 applies to $F=\eta$ with $K_F=(1+r)Cf_F^\theta$ and to $G=\eta\chi_D$ with $K_G=(1+r)\tau(2d)Cf_G^\theta$, so $K_FK_G\le p^{\theta(1+\alpha_1)+o(1)}$. Let $C_0=p^\varepsilon$. For $c\le C_0$ both X/c and Y/c

are $\geq p^{\beta-\varepsilon}$, and $(p^{\beta-\varepsilon})^\kappa \geq K_F K_G p^{\kappa\varepsilon}$ for large p because $\beta > \varpi(1+\alpha_1)+2\varepsilon$. Lemma 4.5(a) gives $|\Phi_\eta(X/c)-\Phi_\eta(Y/c)| \leq 2(2r+3)p^{-\kappa\varepsilon/2}$: the constant $L(1,F)L(1,G)$ cancels. For $c > C_0$ we use $|\Phi_\eta(y)| \leq \sum_{m\leq y}\tau(m)/m \leq (1+\log y)^2$ and $\sum_{c>C_0}|h(c)|/c \leq C_0^{-1/4}\sum_c |h(c)|c^{-3/4} = p^{-\varepsilon/4+o(1)}$. Altogether $S_\eta(u) = O(p^{-\delta})$.

(4) $\eta = 1$. Apply Lemma 4.5(b) with $G = \chi_D$ (primitive of conductor $4up$, $K_G = (1+r)C(4up)^\theta$), $z = p^\varepsilon$ and $y \in \{X/c, Y/c\}$, $c \leq C_0$. Then $K_G(z/y)^\kappa \leq p^{\kappa(2\varepsilon-\beta)+\theta(1+\alpha_1)+o(1)} \leq p^{-\kappa\varepsilon+o(1)}$, so $\Phi_1(X/c)-\Phi_1(Y/c) = L(1,\chi_D)\log(X/Y)+O(p^{-\kappa\varepsilon/2})$: the constants $L(1,\chi_D)\gamma_E + L'(1,\chi_D)$ cancel in the difference. Summing over $c \leq C_0$ with weights $h(c)/c$, using $\sum_{c\leq C_0} h(c)/c = H_u + O(p^{-\varepsilon/4+o(1)})$, $L(1,\chi_D) \ll \log p$, and the bound for $c > C_0$ from (3), gives the claim. □

4.4. **An average of $L(1,\chi_{-4up})$.**

Proposition 4.7. *Let $0 < \alpha_0 < \alpha_1 < 1$ with $\alpha_0 > \varpi^2(1+\alpha_1)+3\varepsilon$. Then*

$$\sum_{u\in\mathcal{U}} \frac{L(1,\chi_{-4up})}{u} \geq \frac{2}{\pi^2}(\alpha_1-\alpha_0)\log p - O(1).$$

Proof. Put $\nu = \varpi(1+\alpha_1)+\varepsilon$ and $N = p^\nu$. For odd n and $u \in \mathcal{U}$, $\chi_{-4up}(n) = \chi_{-4p}(n)\left(\frac{u}{n}\right)$ (reciprocity, $u \equiv 1$ (4)), and $\chi_{-4up}(n) = 0$ for even n. By Lemma 4.4 (conductor $4up$),

$$L(1,\chi_{-4up}) = \sum_{\substack{n\leq N\\ n \text{ odd}}} \frac{\chi_{-4p}(n)}{n}\left(\frac{u}{n}\right) + \varrho_u, \qquad |\varrho_u| \leq (1+r)C(4up)^\theta N^{-\kappa}.$$

Since $\kappa\nu = \theta(1+\alpha_1)+\kappa\varepsilon$, we get $\sum_{u\in\mathcal{U}}|\varrho_u|/u \leq p^{-\kappa\varepsilon+o(1)}$. In the main part exchange the sums: it equals $\sum_{n\leq N \text{ odd}} \chi_{-4p}(n)V(n)/n$ with $V(n) = \sum_{u\in\mathcal{U}}\left(\frac{u}{n}\right)/u$.

If $n = m^2$ then $\left(\frac{u}{m^2}\right) = \mathbf{1}[\gcd(u,m)=1] \geq 0$ and $\chi_{-4p}(m^2) \in \{0,1\}$, so these terms are nonnegative and we keep only $n = 1$. Squarefree $u \equiv 1$ (mod 4) have counting function $\frac{2}{\pi^2}x + O(\sqrt{x})$, so $V(1) = \frac{2}{\pi^2}(\alpha_1-\alpha_0)\log p + O(1)$.

If n is not a square, write $n = n_1 m^2$ with $n_1 > 1$ squarefree. For squarefree u, $\mathbf{1}[u \equiv 1\ (4)]\left(\frac{u}{n}\right) = \frac{1}{2}(\psi_1(u)+\psi_2(u))$ with $\psi_1 = \left(\frac{\cdot}{n_1}\right)\mathbf{1}[\gcd(\cdot,2m)=1]$ and $\psi_2 = \chi_{-4}\left(\frac{\cdot}{n_1}\right)\mathbf{1}[\gcd(\cdot,m)=1]$, completely multiplicative with primitive parts of cube-free conductors n_1, $4n_1 > 1$. Writing $\mu^2(u) = \sum_{e^2|u}\mu(e)$ and using the partial-sum bound of Lemma 4.4,

$$\Big|\sum_{u\leq t}\mu^2(u)\psi_i(u)\Big| \leq \sum_{e\leq\sqrt{t}}\Big|\sum_{v\leq t/e^2}\psi_i(v)\Big| \leq \zeta(2-2\kappa)\tau(2m)C(4n_1)^\theta t^{1-\kappa}$$

($r \geq 3$ gives $2-2\kappa > 1$). Partial summation over $(U_0,U_1]$ gives $|V(n)| \leq (2+r)\zeta(2-2\kappa)\tau(2m)C(4n)^\theta U_0^{-\kappa}$. Hence $\sum_{n\leq N \text{ non-square}}|V(n)|/n \leq p^{o(1)}U_0^{-\kappa}N^\theta = p^{\theta\nu-\kappa\alpha_0+o(1)}$, and $\theta\nu-\kappa\alpha_0 < 0$ because $\alpha_0 > \varpi\nu = \varpi^2(1+\alpha_1)+\varpi\varepsilon$. □

4.5. **Proof of Theorem B.**

Theorem 4.8. *Let $r \geq 3$ be an integer, $\varepsilon \in (0, \frac{1}{100})$, $\varpi = \frac{r+1}{4r}+r\varepsilon$, and let $0 < \alpha_0 < \alpha_1$, $\beta < \gamma$ satisfy*

$$\textit{(C1)}\ \gamma < 1-\alpha_1, \qquad \textit{(C2)}\ \beta > \varpi(1+\alpha_1)+4\varepsilon, \qquad \textit{(C3)}\ \alpha_0 > \varpi^2(1+\alpha_1)+3\varepsilon.$$

Then $M_{\text{gen}}(p) \geq (c-o(1))(\log p)^2$ as $p \to \infty$ through primes $p \equiv 1$ (mod 4), where $c = \frac{8}{\pi^4}(\gamma-\beta)(\alpha_1-\alpha_0)$, and the $o(1)$ is effective.

Proof. For $u \in \mathcal{U}$ and $w \in \mathcal{S}_u \cap (Y, X]$, Lemma 4.2 applies: $8uw \le 8p^{\alpha_1+\gamma} \le p$ for large p by (C1), and $p \nmid uw$. Distinct (u, w) give distinct pairs (the odd entry is u). Since all terms of M_{gen} are nonnegative,

$$M_{\mathrm{gen}}(p) \ge \sum_{u \in \mathcal{U}} \frac{1}{\varphi(u)} \sum_{\eta \in \mathfrak{E}_u} \varepsilon_\eta S_\eta(u) =: M_{\mathfrak{F}}(p).$$

By Proposition 4.6, $|\mathfrak{E}_u| = 2^{\omega(u)+1}$ and $\sum_{u \le U_1} 2^{\omega(u)+1}/\varphi(u) \ll (\log p)^2$,

$$M_{\mathfrak{F}}(p) = (\gamma - \beta) \log p \sum_{u \in \mathcal{U}} \frac{H_u L(1, \chi_{-4up})}{\varphi(u)} + O(p^{-\delta} (\log p)^2).$$

For squarefree odd u and $p \nmid u$, $H_u/\varphi(u) = \frac{4}{\pi^2} \frac{p}{p+1} \cdot \frac{1}{u} \prod_{\ell \mid u} (1 - \ell^{-2})^{-1} \ge \frac{4}{\pi^2} \frac{p}{p+1} \cdot \frac{1}{u}$, and $L(1, \chi_{-4up}) > 0$. Proposition 4.7 gives $M_{\mathfrak{F}}(p) \ge (\gamma - \beta) \frac{4}{\pi^2} \frac{2}{\pi^2} (\alpha_1 - \alpha_0) (\log p)^2 - O(\log p)$. Every constant is effective; positivity of $L(1, \chi)$ is used only through its sign. □

Proof of Theorem B. As $r \to \infty$ and $\varepsilon \to 0$ we have $\varpi \to \frac{1}{4}$, and with the margins removed the conditions read $\gamma < 1 - \alpha_1$, $\beta > \frac{1+\alpha_1}{4}$, $\alpha_0 > \frac{1+\alpha_1}{16}$. Then $(\gamma - \beta)(\alpha_1 - \alpha_0) \to \frac{(3-5\alpha_1)(15\alpha_1 - 1)}{64}$, which is maximal at $\alpha_1 = \frac{1}{3}$ with value $\frac{1}{12}$. Given $\varepsilon' > 0$, fix r, ε and admissible exponents with $\frac{8}{\pi^4}(\gamma - \beta)(\alpha_1 - \alpha_0) > \frac{2}{3\pi^4} - \varepsilon'/2$, and apply Theorem 4.8. □

Remark 4.9. (1) With $r = 3$ alone ($\varpi = \frac{1}{3}$) the optimum is at $\alpha_1 = \frac{5}{16}$ and gives the constant $\frac{1}{3\pi^4}$. Under a fixed zero-free strip for the relevant L-functions one could let $\beta, \alpha_0 \to 0$ and obtain $\frac{2}{\pi^4}$; we do not use this. The constant is not optimised: we discard the squares $n = m^2 > 1$ in Proposition 4.7, the factor $\prod_{\ell \mid u} (1 - \ell^{-2})^{-1}$, all $u \equiv 3 \pmod 4$ and all $B \equiv 0 \pmod 4$.

(2) *The step that fails at odd squares.* $M_{\mathrm{gen}}(m^2) = 0$ for odd m, so the argument must break if p is replaced by m^2. The decisive place is step (2) of Proposition 4.6. With p replaced by m^2 and $\gcd(m, 2u) = 1$, χ_{-4um^2} agrees with χ_{-4u} on integers prime to m, and the character $\eta_{u,1}\chi_{-4um^2}$ is principal on its support. In fact every w counted by ρ_u has $\eta_{u,1}(w) = \chi_{-4u}(w) = 1$, so $S_{\eta_{u,1}} = S_1$, while $\varepsilon_{u,1} = \left(\frac{-2}{u}\right) \chi_{-4}(-(m^2 + u)/2) = -1$ for $u \equiv 1$ and for $u \equiv 5 \pmod 8$ (as $m^2 \equiv 1 \pmod 8$). So the principal term is cancelled exactly. At a prime p this cannot happen because p divides the conductor of $\eta\chi_{-4up}$.

(3) The proof never evaluates an L-function at a single small modulus and never averages over p: Proposition 4.6 averages over w at fixed u and p, and Proposition 4.7 averages over u at fixed p.

4.6. **The genus layer of a slice.** For odd u let $S_u(p) := \sum_{B:\,(u,B) \in \mathcal{P}(p)} \frac{s(q)}{\varphi(q)} \mathbf{1}[pa \in ((\mathbb{Z}/q\mathbb{Z})^\times)^2]$ ($q = 4uB$, $a = -(u + B)$); every pair in $\mathcal{P}(p)$ has exactly one even entry and the summand is symmetric, so $M_{\mathrm{gen}}(p) = 2 \sum_{u \text{ odd}} S_u(p)$.

Proposition 4.10. *Let $u \equiv 1 \pmod 4$ be squarefree with $u \le p^\alpha$, where $1 - \alpha > \varpi(1 + \alpha) + 5\varepsilon$ (for suitable r, ε this allows every fixed $\alpha < \frac{3}{5}$). Put $D = -4up$ and $X_u = \lfloor (p + u)/(2(4u - 1)) \rfloor$. Then, with $\delta = \delta(r, \varepsilon, \alpha) > 0$,*

$$S_u(p) = \frac{1}{2\varphi(u)} \sum_{\eta \in \mathfrak{E}_u} \varepsilon_\eta \Sigma_\eta + O(p^{-\delta}),$$

where, with $H_{u,\eta}(s) := \prod_\ell h_\ell(\eta(\ell)\ell^{-s})$, $h_\ell(Y) = 1 - Y$ *for* $\ell \mid 2up$, $h_\ell(Y) = 1 + \frac{2Y}{\ell-1} - \frac{(\ell+1)Y^2}{\ell-1}$ *if* $\chi_D(\ell) = 1$, $h_\ell(Y) = 1 - Y^2$ *if* $\chi_D(\ell) = -1$, *and* $H_u = H_{u,1}$,

$$\Sigma_1 = L(1,\chi_D)H_u(1)(\log X_u + \gamma_E) + L'(1,\chi_D)H_u(1) + L(1,\chi_D)H_u'(1),$$
$$\Sigma_\eta = L(1,\eta)\,L(1,\eta\chi_D)\,H_{u,\eta}(1) \qquad (\eta \neq 1).$$

For $\eta = \eta_{d,j} \neq 1$, $L(1,\eta) = L(1,\chi_{d^*})$ *(*$j = 0$*) or* $L(1,\chi_{-4d^*})$ *(*$j = 1$*), where* $d^* = (-1)^{(d-1)/2}d$, *and* $L(1,\eta\chi_D) = L(1,\chi_{D'})\prod_{\ell\mid 2d}(1-\chi_{D'}(\ell)/\ell)$ *with* $D' = -4(ep)^*$ *(*$j = 0$*) or* $D' = (ep)^*$ *(*$j = 1$*),* $e = u/d$.

Proof. In the slice, $u + B$ is odd, so B is even. If $4 \mid B$ then $8 \mid q$ and the genus condition forces $a \equiv p \pmod 8$, hence $-u \equiv p \equiv 1 \pmod 4$, impossible; so $B = 2w$ with w odd. The condition $4uB - u - B \le p$ is $w \le X_u$, and $X_u < p$ so $p \nmid w$; and $\gcd(w,u) = 1$. The proof of Lemma 4.2 did not use that w is squarefree (for $\ell^k \parallel w$ the group $(\mathbb{Z}/\ell^k)^\times$ is cyclic, and a unit is a square modulo ℓ^k iff modulo ℓ), so

$$S_u(p) = \frac{1}{2\varphi(u)}\sum_{\eta\in\mathfrak{E}_u}\varepsilon_\eta\Sigma_\eta(X_u), \qquad \Sigma_\eta(X) := \sum_{\substack{w\le X\\ \gcd(w,2up)=1}} \eta(w)\frac{2^{\omega(w)}}{\varphi(w)}\mathbf{1}[\chi_D(\ell) = 1\ \forall \ell \mid w].$$

The summand is $\eta(w)g(w)/w$ with g multiplicative, supported on integers prime to $2up$, with local series $1 + \frac{2\ell}{\ell-1}\frac{Y}{1-Y}$ at split ℓ and 1 at inert ℓ. So $g = 1 * \chi_D * h$ with $\sum_k h(\ell^k)Y^k = h_\ell(Y)$ as stated (the split factor is $(1 + \frac{2\ell}{\ell-1}\frac{Y}{1-Y})(1-Y)^2$), all $|h(\ell^k)| \le 2$ and $h(\ell^k) = 0$ for $k \ge 3$; hence $\sum_c |h(c)|c^{-3/4}(1 + \log c) = p^{o(1)}$. As before $\Sigma_\eta(X) = \sum_c \frac{\eta(c)h(c)}{c}\Phi_\eta(X/c)$. Put $y_0 = (4up)^{\varpi}p^{3\varepsilon}$. For $c \le X/y_0$, Lemma 4.5 gives $\Phi_\eta(X/c) = L(1,\eta)L(1,\eta\chi_D) + O(p^{-\kappa\varepsilon})$ for $\eta \neq 1$ (part (a); $K_F K_G \le (4up)^\theta p^{o(1)}$), and $\Phi_1(X/c) = L(1,\chi_D)(\log(X/c) + \gamma_E) + L'(1,\chi_D) + O(p^{-\kappa\varepsilon})$ (part (b) with $z = p^\varepsilon$). The terms with $c > X/y_0$, and the tails $c > X/y_0$ of the c-sums of the main terms, are $\ll (X/y_0)^{-1/4}p^{o(1)}(\log p)^3$, using $|\Phi_\eta(y)| \le (1 + \log y)^2$ and $|L(1,\cdot)| \ll \log p$, $|L'(1,\cdot)| \ll (\log p)^2$ for the characters involved (conductors $\le 4up$). Finally $X_u/y_0 \ge p^{1-\alpha-\varpi(1+\alpha)-3\varepsilon}/32 \ge p^\varepsilon$ for large p, and $\sum_c \frac{h(c)}{c}(\log X - \log c) = H_u(1)\log X + H_u'(1)$. The identification of $L(1,\eta)$ and $L(1,\eta\chi_D)$ follows from step (2) of Proposition 4.6 and $\left(\frac{n}{d}\right) = \left(\frac{d^*}{n}\right)$ for odd $n > 0$. □

Remark 4.11. Proposition 4.10 writes the genus layer of each slice $u \equiv 1 \pmod 4$ squarefree, $u \le p^\alpha$, as a principal "class-number" term ($L(1,\chi_{-4up}) = \pi h(-4up)/\sqrt{4up}$, times a logarithm) plus $2^{\omega(u)+1} - 1$ cross terms of either sign, each a product of two quadratic L-values whose second factor has p in its conductor. For $d = u$, $j = 1$ the second factor is $L(1,\chi_p)\cdot(1 - \chi_p(2)/2)$ up to Euler factors. The cross terms can be large and negative for individual slices; the windows of Theorem 4.8 are chosen so that their constant main terms cancel. The additive terms $L'(1,\chi_D)H_u(1)$ and $L(1,\chi_D)H_u'(1)$ in Σ_1 are kept: we have no bound making them uniformly $o(L(1,\chi_D)H_u(1)\log X_u)$ as u and p vary. We have checked the formula numerically against the exact slice sums (see §6). We do not give a closed form for slices with $u \equiv 3 \pmod 4$, non-squarefree u, or $u > p^{3/5}$.

4.7. **Characters of order at least three modulo** $8u$**.** We keep the windows and the family of Theorem 4.8: $\mathfrak{F} = \mathfrak{F}(p) = \{(u,w) : u \in \mathcal{U},\ w \in \mathcal{S}_u \cap (Y, X]\}$, each (u,w) standing for the two pairs $(u,2w), (2w,u)$ with $q = 8uw$, $a = -(u + 2w)$. By the Chinese remainder theorem, $\mathbf{1}[p \equiv a\ (q)] = A_u(w)B_u(w)$ with

$$A_u(w) := \mathbf{1}[p \equiv a \pmod{8u}], \qquad B_u(w) := \mathbf{1}[p \equiv a \pmod w] = \mathbf{1}[w \mid p + u].$$

The real-character (genus) parts of these two indicators, i.e. their projections onto the real characters modulo $8u$ and modulo w, are

$$A_u^{\mathrm{gen}}(w)=\frac{1}{2\varphi(u)}\sum_{\eta\in\mathfrak{E}_u}\varepsilon_\eta\eta(w),\qquad B_u^{\mathrm{gen}}(w)=\rho_u(w)$$

(Lemma 4.2 and its proof). Thus the subfamily's contributions to $R(p)$ and to $M_{\mathrm{gen}}(p)$ are $R_{\mathfrak{F}}=2\sum_{\mathfrak{F}}A_uB_u$ and $M_{\mathfrak{F}}=2\sum_{\mathfrak{F}}A_u^{\mathrm{gen}}\rho_u$. Define the intermediate quantity

$$M_{\mathfrak{F}}^{\mathrm{cls}}(p):=2\sum_{(u,w)\in\mathfrak{F}}A_u(w)\rho_u(w),$$

in which the condition modulo $8u$ is the exact class condition (all characters modulo $8u$, of every order), while the condition modulo w is replaced by its genus density.

Theorem 4.12. *Let $r\ge1$ be an integer, $\varepsilon\in(0,\frac{1}{100})$, $\varpi=\frac{r+1}{4r}+r\varepsilon$, and assume*

$$\text{(C1) }\gamma<1-\alpha_1,\qquad \text{(C2*) }\beta>2\alpha_1+\varpi+4\varepsilon,\qquad \text{(C4) }\alpha_1<\tfrac15.$$

There is $\delta>0$ such that, uniformly in $u\in\mathcal{U}$ and in the unit class $s \bmod 4u$,

$$\sum_{\substack{w\in\mathcal{S}_u\cap(Y,X]\\ w\equiv s\,(4u)}}\rho_u(w)=\mathfrak{M}_u+O(p^{-\delta}/u),\tag{10}$$

with $\mathfrak{M}_u$ independent of s. Consequently $M_{\mathfrak{F}}^{\mathrm{cls}}(p)-M_{\mathfrak{F}}(p)=O(p^{-\delta}\log p)$. If moreover $r\ge3$ and (C3) of Theorem 4.8 holds, then $M_{\mathfrak{F}}^{\mathrm{cls}}(p)\ge(c-o(1))(\log p)^2$ with $c=\frac{8}{\pi^4}(\gamma-\beta)(\alpha_1-\alpha_0)$; admissible windows give every constant below $c_X=\frac{121}{640\pi^4}=0.00194\ldots$. All constants are effective.

Only Burgess's bound for the Legendre symbol modulo the prime p is used in the proof of (10). The theorem says that on $\mathfrak{F}$ the characters of order ≥3 modulo $8u$, weighted by the genus density at w, carry no main term. It says nothing about the exact divisibility $w\mid p+u$, which is where the rest of $R_{\mathfrak{F}}-M_{\mathfrak{F}}$ lives.

Proof. Let $q=4u$, s a unit modulo q, and h as in Proposition 4.6(1), so that $\rho_u(w)=\sum_{abc=w}\frac1a\frac{\chi_D(b)}{b}\frac{h(c)}{c}$. As $w\equiv s$ forces $\gcd(abc,q)=1$,

$$\sum_{w\equiv s}\rho_u(w)=\sum_{\gcd(c,q)=1}\frac{h(c)}{c}\big[\Phi_{s\bar c}(X/c)-\Phi_{s\bar c}(Y/c)\big],\qquad \Phi_t(y):=\sum_{\substack{ab\le y\\ ab\equiv t\,(q)}}\frac{\chi_D(b)}{ab}.$$

Fix $\theta'=\frac12-\varepsilon$, so $\sum_c|h(c)|c^{\theta'-1}=p^{o(1)}$, and let $C_0=p^{\varepsilon}$. We use the elementary estimate

$$\Big|\sum_{\substack{T_1<a\le T_2\\ a\equiv v\,(q)}}\frac1a-\frac1q\log\frac{T_2}{T_1}\Big|\le\frac{1}{T_1}\qquad(T_1>0),\tag{11}$$

which follows by comparing the sum with $\frac1q\int dt/t$.

(i) Large c. For $C_0<c\le Y/q$, the bracket is at most $\sum_{Y/c<m\le X/c,\,m\equiv t}\tau(m)/m\le p^{o(1)}(c/Y+q^{-1}\log(X/Y))\le p^{o(1)}/u$, and $\sum_{c>C_0}|h(c)|/c\le C_0^{-\theta'}p^{o(1)}$. For $c>Y/q$ the bracket is $\le(1+\log X)^2$ and $\sum_{c>Y/q}|h(c)|/c\le(Y/q)^{-\theta'}p^{o(1)}=p^{-\theta'(\beta-\alpha_1)+o(1)}$. By (C2*) and $\varpi\ge\frac14$, $\beta-\alpha_1>\alpha_1+\frac14+4\varepsilon$, and by (C4) and $\varepsilon<\frac{1}{100}$, $(\frac12-\varepsilon)(\alpha_1+\frac14+4\varepsilon)\ge\alpha_1+\frac{\varepsilon}{2}$. So both ranges contribute $O(p^{-\min(\varepsilon\theta',\varepsilon/2)+o(1)}/u)$.

(ii) Small c. Let $c \le C_0$ and $B_0 := qp^{\varpi+\varepsilon}$. For $y \ge 1$ we have the exact identity

$$\Phi_t(y) = \sum_{b \le B_0} \frac{\chi_D(b)}{b}\, \alpha_{t\bar b}(y/b) + \sum_{a \le y/B_0} \frac{1}{a}\big[\beta_{t\bar a}(y/a) - \beta_{t\bar a}(B_0)\big],$$

where $\alpha_v(T) = \sum_{a \le T,\, a \equiv v} 1/a$ and $\beta_v(T) = \sum_{b \le T,\, b \equiv v} \chi_D(b)/b$ (if $\chi_D(b) \ne 0$ then b is a unit modulo q, and then so is a).

Second part. For a unit v modulo q and $b \equiv v$, b odd and prime to u, so $\chi_D(b) = \chi_{-4}(b)\left(\frac{b}{u}\right)\left(\frac{b}{p}\right) = \chi_{-4}(v)\left(\frac{v}{u}\right)\left(\frac{b}{p}\right)$. Writing $b = v_0 + qk$ $(1 \le v_0 \le q)$, $\left(\frac{b}{p}\right) = \left(\frac{q}{p}\right)\left(\frac{k+v_0\bar q}{p}\right)$, where $\bar q$ is the inverse of q modulo p. Burgess's bound for the Legendre symbol modulo p, with r and ε, bounds every sum of $\left(\frac{k+v_0\bar q}{p}\right)$ over K consecutive k by $CK^{1-1/r}p^{\varpi/r}$. Partial summation against the decreasing weight $1/(v_0+qk)$, for k beyond $k_0 \ge B_0/q - 1$, gives $|\beta_v(T) - \beta_v(B_0)| \le (r+2)Cp^{\varpi/r}k_0^{-1/r}/q \le 2(r+2)Cp^{-\varepsilon/r}/q$. Hence the second part is $O(p^{-\varepsilon/r+o(1)}/u)$ for $y \in \{X/c, Y/c\}$.

First part. By (11) with $T_1 = Y/(cb) \ge Y/(C_0B_0) \ge 1$,

$$\sum_{b \le B_0} \frac{\chi_D(b)}{b}\Big[\alpha_{t\bar b}\Big(\frac{X}{cb}\Big) - \alpha_{t\bar b}\Big(\frac{Y}{cb}\Big)\Big] = \frac{1}{q}\log\frac{X}{Y}\cdot\Lambda(B_0) + O\Big(\frac{cB_0}{Y}\Big), \quad \Lambda(B_0) := \sum_{b \le B_0}\frac{\chi_D(b)}{b},$$

and $cB_0/Y \le 4u\,p^{2\varepsilon+\varpi-\beta} \ll p^{-\varepsilon}/u$ by (C2*). The main term does not depend on t. Summing over $c \le C_0$ we obtain (10) with $\mathfrak{M}_u = q^{-1}\log(X/Y)\Lambda(B_0)\sum_{c \le C_0,\,(c,q)=1} h(c)/c$ and any $\delta < \min(\varepsilon/r, \varepsilon\theta', \varepsilon/2)$.

(iii) Consequences. On $\mathcal{S}_u$, $A_u(w) = \mathbf{1}[w \equiv s_u \ (4u)]$ for the unit class s_u determined by $w \equiv w_0 \pmod 4$ and $2w \equiv -p \pmod u$. By the proof of Lemma 4.2, $A_u^{\mathrm{gen}}(w) = |\mathcal{C}|^{-1}\mathbf{1}[w \in \mathcal{C}]$ with $\mathcal{C} = s_u((\mathbb{Z}/4u\mathbb{Z})^\times)^2$, $|\mathcal{C}| = \varphi(u)/2^{\omega(u)}$. By (10), $\sum_w (A_u - A_u^{\mathrm{gen}})(w)\rho_u(w) = O(p^{-\delta}/u)$; summing over $u \le U_1$ gives $M_{\mathfrak{F}}^{\mathrm{cls}} - M_{\mathfrak{F}} = O(p^{-\delta}\log p)$. Since $\varpi\alpha_1 \le 2\alpha_1$, (C2*) implies (C2) of Theorem 4.8, and with (C3) the proof of that theorem gives $M_{\mathfrak{F}} \ge (c - o(1))(\log p)^2$. Letting $r \to \infty$, $\varepsilon \to 0$, $\gamma \uparrow 1-\alpha_1$, $\beta \downarrow 2\alpha_1 + \frac14$, $\alpha_0 \downarrow \frac{1+\alpha_1}{16}$, the product $(\gamma-\beta)(\alpha_1-\alpha_0)$ tends to $\frac{(3/4-3\alpha_1)(15\alpha_1-1)}{16}$, maximal at $\alpha_1 = \frac{19}{120} < \frac15$ with value $\frac{121}{5120}$. □

5. Almost all primes

5.1. **A truncated count.** Let X be large and $L = \log X$. Fix $A_0 = 200$ and let $B_0 = B(A_0)$ be the exponent in the Bombieri–Vinogradov theorem below. Put

$$Q := X^{1/4}L^{-B_0/2}, \qquad Y := \frac{Q+1}{4},$$

$$\mathcal{T} := \{(u,C) \in \mathbb{N}^2 :\ m_{uC} := 4uC - 1 \le Q\} = \{(u,C) : uC \le Y\}.$$

For $n \in \mathbb{N}$ define the weighted truncated count

$$N(n) := \sum_{(u,C)\in\mathcal{T}} c_{uC}\,\mathbf{1}[m_{uC} \mid n+u], \qquad c_{uC} := \frac{\varphi(m_{uC})}{m_{uC}} \in (0,1].$$

Since $\gcd(u, 4uC-1) = 1$, the event $m \mid n+u$ is $n \equiv -u \pmod m$ with $-u$ a unit modulo m.

Lemma 5.1. *If p is prime and $p \ge (Q+1)^2/4$, then $R(p) \ge 2N(p)$.*

Proof. If $(u,C) \in \mathcal{T}$ and $m = 4uC-1 \mid p+u$, then $B := (p+u)/m \in \mathbb{N}$ satisfies $4uBC - u - B = Bm - u = p$, so (u,B,C) is a triple counted by $R(p)$ with $4uC - 1 \le Q$, and different (u,C) give different triples. Applying the involution of Lemma 2.1(ii) gives as many triples with $4BC - 1 \le Q$. No triple has both: $4uC \le Q+1$ and $4BC \le Q+1$ would give $p < 4uBC \le (4uC)(4BC)/4 \le (Q+1)^2/4$. Since $c_{uC} \le 1$, the claim follows. □

Lemma 5.2. $M := \sum_{(u,C)\in\mathcal{T}} c_{uC}/\varphi(m_{uC}) = \sum_{uC\le Y} \frac{1}{4uC-1} \ge \frac{1}{8}\log^2 Y$.

Proof. $1/(4uC-1) > 1/(4uC)$ and $\sum_{C\le Y/u} 1/C \ge \log(Y/u)$, so $M \ge \frac14 \sum_{u\le Y} \log(Y/u)/u$. The function $t \mapsto \log(Y/t)/t$ is decreasing and nonnegative on $[1,Y]$, so the sum is $\ge \int_1^Y \log(Y/t)\,dt/t = \frac12\log^2 Y$. □

5.2. **First and second moments.**

Cited theorem 5.3 (Bombieri–Vinogradov [1, 21]; [13, Thm. 17.1], [6, Ch. 28])**.** For every $A>0$ there is $B=B(A)$ such that

$$\sum_{q\le x^{1/2}(\log x)^{-B}} \max_{y\le x} \max_{\gcd(a,q)=1} \Big|\pi(y;q,a) - \frac{\pi(y)}{\varphi(q)}\Big| \ll_A \frac{x}{(\log x)^A}.$$

(The standard statement is for $\psi(y;q,a)$; the form for π follows by partial summation, at the cost of enlarging B. The implied constant is ineffective.) Write $E(X;q,a) := \pi(X;q,a) - \pi(X)/\varphi(q)$; trivially $|E(X;q,a)| \le 3X/\varphi(q)$ for $q\le X$. We use the classical bounds $\sum_{m\le x}\tau(m)^k \ll_k x(\log x)^{2^k-1}$, $m/\varphi(m) \ll \log\log m$, $\tau(ab)\le\tau(a)\tau(b)$ and $\varphi(ab)\ge\varphi(a)\varphi(b)$. Let $r(m) := \#\{(u,C): 4uC-1=m\} \le \tau(m+1)$. Note $Q^2 = X^{1/2}L^{-B_0}$.

Lemma 5.4. $S_1 := \sum_{p\le X} N(p) = \pi(X)M + O(XL^{-90})$.

Proof. We have $S_1 = \sum_{(u,C)\in\mathcal{T}} c_{uC}\pi(X;m,-u) = \pi(X)M + \sum c_{uC}E(X;m,-u)$. The error is at most $\sum_{m\le Q} r(m)\max_a|E(X;m,a)|$, which by Cauchy–Schwarz is at most

$$\Big(\sum_m r(m)^2\frac{3X}{\varphi(m)}\Big)^{1/2}\Big(\sum_{m\le Q}\max_a|E(X;m,a)|\Big)^{1/2} \ll (XL^5)^{1/2}(XL^{-200})^{1/2}.$$

□

Call $P=(u,C)$ and $P'=(u',C')$ in $\mathcal{T}$ *compatible* if $u\equiv u' \pmod{\gcd(m_P,m_{P'})}$.

Lemma 5.5. $S_2 := \sum_{p\le X} N(p)^2 = \pi(X)\sum_{P,P' \text{ compatible}} \frac{c_Pc_{P'}}{\varphi(\mathrm{lcm}(m_P,m_{P'}))} + O(XL^{-10})$.

Proof. The system $p\equiv -u\ (m)$, $p\equiv -u'\ (m')$ is solvable iff P,P' are compatible, and then it is a single unit class modulo $\mathrm{lcm}(m,m') \le Q^2$. Let $\nu(k) := \#\{(P,P') : \mathrm{lcm}(m_P,m_{P'})=k\} \le \varrho(k)^2$ with $\varrho(k)=\sum_{m|k} r(m)$. By Hölder, $\varrho(k)^4 \le \tau(k)^3\sum_{m|k}\tau(m+1)^4$, so

$$\sum_{k\le Q^2}\frac{\varrho(k)^4}{\varphi(k)} \le \sum_{m\le Q^2}\frac{\tau(m+1)^4\tau(m)^3}{\varphi(m)}\sum_{j\le Q^2}\frac{\tau(j)^3}{\varphi(j)} \ll L^{161}\cdot L^9,$$

using $\sum_{m\le x}\tau(m+1)^4\tau(m)^3 \le (\sum\tau(m+1)^8)^{1/2}(\sum\tau(m)^6)^{1/2} \ll x(\log x)^{159}$ and partial summation. By Cauchy–Schwarz as in Lemma 5.4, the total error is $\ll (XL^{170})^{1/2}(XL^{-200})^{1/2} = XL^{-15}$. □

5.3. **The variance.**

Lemma 5.6. *Let $(q_P,a_P,\beta_P)_P$ be a finite family with $\gcd(a_P,q_P)=1$ and $\beta_P\ge 0$. For a primitive character ψ of conductor d put $S_\psi = \sum_{P:\,d\mid q_P}\beta_P\bar\psi(a_P)/\varphi(q_P)$. Then*

$$\sum_{\substack{P,P'\\ a_P\equiv a_{P'}\ (\gcd(q_P,q_{P'}))}} \frac{\beta_P\beta_{P'}}{\varphi(\mathrm{lcm}(q_P,q_{P'}))} = \sum_\psi |S_\psi|^2,$$

the sum running over all primitive characters, including $\psi=1$.

Proof. $\varphi(\mathrm{lcm}(q,q'))\varphi(\gcd(q,q')) = \varphi(q)\varphi(q')$, so the left side is

$$\sum_{P,P'} \frac{\beta\beta'\varphi(g)\mathbf{1}[a \equiv a' \ (g)]}{\varphi(q)\varphi(q')}, \qquad g = \gcd(q,q').$$

By orthogonality $\varphi(g)\mathbf{1}[a \equiv a' \ (g)] = \sum_{\chi \bmod g} \bar\chi(a)\chi(a')$, and each $\chi \bmod g$ is induced by a unique primitive ψ of conductor $d \mid g$, with $\chi(a) = \psi(a)$ for a prime to g. Exchange the sums; $d \mid g$ iff $d \mid q$ and $d \mid q'$. □

With $q_P = m_P$, $a_P = -u$, $\beta_P = c_P$ we have $\beta_P/\varphi(q_P) = 1/m_P$, the term $\psi = 1$ equals M^2, and by Lemmas 5.4, 5.5, 5.6,

$$\sum_{p \le X} (N(p) - M)^2 = S_2 - 2MS_1 + M^2\pi(X) = \pi(X)\,V + O(XL^{-8}), \qquad V := \sum_{d>1} \sideset{}{^*}\sum_{\psi \bmod d} |S_\psi|^2, \tag{12}$$

where $S_\psi = \sum_{(u,C)\in\mathcal{T}:\, d \mid m_{uC}} \bar\psi(-u)/m_{uC}$ and $\sum^*$ is over primitive characters.

Lemma 5.7. *For $d > 1$ and $r \in (\mathbb{Z}/d\mathbb{Z})^\times$ let $A_r^{(d)} := \sum_{(u,C)\in\mathcal{T}:\, d \mid 4uC-1,\ u \equiv -r\,(d)} \frac{1}{4uC-1}$. Then for any real numbers Λ_d, $V \le \sum_{1<d\le Q} \varphi(d) \sum_r (A_r^{(d)} - \Lambda_d)^2$.*

Proof. For $d > 1$ primitive characters are non-principal, so $\sum_\psi^* |S_\psi|^2 \le \sum_{\chi \ne \chi_0} |T_\chi|^2$ with $T_\chi = \sum_r \bar\chi(r) A_r^{(d)}$. By orthogonality $\sum_\chi |T_\chi|^2 = \varphi(d)\sum_r (A_r^{(d)})^2$, and removing χ_0 gives $\varphi(d)\sum_r (A_r - \bar A)^2 \le \varphi(d)\sum_r (A_r - \Lambda_d)^2$, $\bar A$ the mean. Terms with $d > Q$ vanish. □

The one-cell structure. If $d \mid 4uC - 1$ then d is odd and prime to u and C; for odd $d > 1$ the conditions $u \equiv -r$, $d \mid 4uC - 1$ are equivalent to

$$u \equiv x_r := -r, \qquad C \equiv y_r := -(4r)^{-1} \pmod d.$$

So $A_r^{(d)}$ is a sum over a single lattice cell of the hyperbolic region $uC \le Y$. For even d, $A_r^{(d)} = 0$.

5.4. **Hyperbolic progression sums.** For $d \ge 1$ and $x, y \in \mathbb{Z}$ let $u_0, C_0 \in [1, d]$ be the least positive integers with $u_0 \equiv x$, $C_0 \equiv y \pmod d$.

Lemma 5.8. *For $Z > 0$ let $h(Z) = \sum_{1\le C\le Z,\, C\equiv y\,(d)} 1/C$. Then $|h(Z) - \mathbf{1}[C_0 \le Z]/C_0 - \frac{1}{d}\log^+(Z/d)| \le 2/d$.*

Proof. If $Z < C_0 + d$ then $h(Z) = \mathbf{1}[C_0 \le Z]/C_0$ and $\log^+(Z/d) < \log 2$. Otherwise let $J = \lfloor (Z - C_0)/d \rfloor \ge 1$ and $\vartheta = C_0/d \in (0,1]$; then $h(Z) = 1/C_0 + \frac{1}{d}\sum_{j=1}^J (j+\vartheta)^{-1}$, and comparison with integrals puts $\sum_{j=1}^J (j+\vartheta)^{-1}$ between $\log((J+1)/2)$ and $1 + \log(J+1)$. Since $\log(J+1) \in [\log(Z/d) - \log 2, \log(Z/d) + \log 2]$, the sum differs from $\log(Z/d)$ by at most $1 + \log 2 < 2$. □

Lemma 5.9. *Let $Y \ge 1$, $d \le 4Y$, $H := \sum_{u\equiv x,\, C\equiv y\,(d);\ uC\le Y} \frac{1}{uC}$, $\Lambda := \frac{1}{2d^2}(\log^+(Y/d^2))^2$ and $\mathcal{L} := \log(4Y) + 2$. Then*

$$|H - \Lambda| \le \frac{1}{u_0 C_0} + \frac{\mathcal{L}}{dC_0} + \frac{\mathcal{L}}{du_0} + \frac{3\mathcal{L}(1+\log d)}{d^2}.$$

Proof. $H = \sum_{u\equiv x,\, u\le Y} u^{-1} h(Y/u)$; inserting Lemma 5.8 gives $H = H_1 + H_2 + H_3$ with $H_1 = C_0^{-1}\sum_{u\equiv x,\, u\le Y/C_0} u^{-1}$, $H_2 = d^{-1}\sum_{u\equiv x,\, u\le Y/d} u^{-1}\log(Y/(ud))$ and $|H_3| \le \frac{2}{d}\sum_{u\equiv x,\, u\le Y} u^{-1}$. By Lemma 5.8 for u, $\sum_{u\equiv x,\, u\le Z} u^{-1} \le u_0^{-1} + (\log^+(Z/d) + 2)/d$, so $|H_1| \le \frac{1}{u_0C_0} + \frac{\log^+(Y/d)+2}{dC_0}$ and $|H_3| \le \frac{2}{du_0} + \frac{2(\log^+(Y/d)+2)}{d^2}$. For H_2 let $g(t) = \log(Y/(td))/t$, nonnegative and non-increasing

on $(0, Y/d]$. If $u_0 > Y/d$ then $H_2 = 0$ and $Y < d^2$, so $\Lambda = 0$. Otherwise $dH_2 = g(u_0) + \sum_{j=1}^{J} g(u_0 + jd)$, $J = \lfloor (Y/d - u_0)/d \rfloor$, and comparison with integrals gives $\frac{1}{2d}(\log^+ \frac{Y}{(u_0+d)d})^2 \le \sum_{j=1}^{J} g(u_0 + jd) \le \frac{1}{2d}\log^2 \frac{Y}{u_0 d}$. With $\lambda = \log^+(Y/d^2)$ and $\delta = \log(d/u_0) \in [0, \log d]$ we have $0 \le \log \frac{Y}{u_0 d} \le \lambda + \delta$ and $\log^+ \frac{Y}{(u_0+d)d} \ge (\lambda - \log 2)^+$; since $d\Lambda = \lambda^2/(2d)$,

$$|dH_2 - g(u_0) - d\Lambda| \le \frac{\lambda\delta + \delta^2/2 + \lambda \log 2}{d}, \qquad 0 \le \frac{g(u_0)}{d} \le \frac{\lambda + \delta}{du_0}.$$

Now $\lambda + \delta \le \log(4Y)$, $\log^+(Y/d) \le \log Y$, and $\lambda\delta + \delta^2/2 + \lambda \log 2 \le \log(4Y)(1.5 \log d + 1)$. Collecting terms gives the claim. □

Proposition 5.10. *$V \le \frac{4}{9}\big[4(1 + \log Q)^2 + 2\zeta(2)\mathcal{L}^2 \log Q + 9K_1\mathcal{L}^2\big] = (\frac{8\zeta(2)}{9} + o(1)) \log^3 Q$, where $\mathcal{L} = \log(Q + 1) + 2$ and $K_1 = \sum_{d \ge 2}(1 + \log d)^2/d^2$.*

Proof. Take $\Lambda_d = 0$ for even d. Fix odd $d \in (1, Q]$ and r, and let $(u_0, C_0) \in [1, d]^2$ represent (x_r, y_r). Note $d \le Q < 4Y$. Since $\frac{1}{4uC-1} = \frac{1}{4uC} + \frac{1}{4uC(4uC-1)}$, we have $A_r^{(d)} = H_r/4 + E_r$ with H_r the sum H of Lemma 5.9 for the cell, and $0 \le E_r \le \frac{1}{12}\sum_{\text{cell}}(uC)^{-2} \le \frac{1}{12}(u_0^{-2} + \zeta(2)d^{-2})(C_0^{-2} + \zeta(2)d^{-2})$. Take $\Lambda_d = \Lambda/4$. Using $\zeta(2) \le \mathcal{L}$ and $\zeta(2)^2 \le 3\mathcal{L}$, Lemma 5.9 gives

$$|A_r^{(d)} - \Lambda_d| \le \frac{1}{3}\Big[\frac{1}{u_0 C_0} + \frac{\mathcal{L}}{dC_0} + \frac{\mathcal{L}}{du_0} + \frac{3\mathcal{L}(1 + \log d)}{d^2}\Big].$$

Square with $(a + b + c + e)^2 \le 4(a^2 + b^2 + c^2 + e^2)$. As r runs over $(\mathbb{Z}/d\mathbb{Z})^\times$, the u_0 are distinct elements of $[1, d]$, and so are the C_0; so $\sum_r u_0^{-2}, \sum_r C_0^{-2} < \zeta(2)$. With $\varphi(d) \le d$ and $\sum_{2 \le d \le Q} 1/d \le \log Q$,

$$V \le \frac{4}{9}\Big[T + 2\zeta(2)\mathcal{L}^2 \log Q + 9K_1\mathcal{L}^2\Big], \qquad T := \sum_{\substack{1 < d \le Q \\ d \text{ odd}}} \varphi(d) \sum_r \frac{1}{(u_0 C_0)^2}.$$

The map $(d, r) \mapsto (u_0, C_0)$ lands in pairs with $d \mid 4u_0C_0 - 1$ and $u_0, C_0 \le Q$, and $r \equiv -u_0$ is determined by (d, u_0). Hence $T \le \sum_{u,C \le Q}(uC)^{-2} \sum_{d \mid 4uC-1} \varphi(d) = \sum_{u,C \le Q} \frac{4uC-1}{(uC)^2} \le 4(1 + \log Q)^2$. □

5.5. **Proof of Theorem C.** By Lemma 5.2, $M \ge \frac{1}{8}\log^2 Y$ with $\log Y = \log Q + O(1)$, and $\log Q = \frac{L}{4} - \frac{B_0}{2}\log L$. So Proposition 5.10 gives

$$\frac{V}{M^2} \le \frac{64 \cdot \frac{8\zeta(2)}{9} + o(1)}{\log Q} = \frac{\frac{2048\zeta(2)}{9} + o(1)}{L} \le \frac{375}{L}$$

for large X ($\frac{2048\zeta(2)}{9} = 374.3\ldots$). Let $\eta \in (0, 1)$. By (12) and Chebyshev's inequality,

$$\#\{p \le X : |N(p) - M| \ge \tfrac{\eta}{2}M\} \le \frac{\pi(X)V + O(XL^{-8})}{(\eta/2)^2 M^2} \le \frac{1500}{\eta^2}\frac{\pi(X)}{L} + O(\eta^{-2}XL^{-12}).$$

Let $p \in (X^{1/2}, X]$ be outside this set. Then $p \ge (Q + 1)^2/4$, and by Lemmas 5.1 and 5.2

$$R(p) \ge 2N(p) \ge \Big(1 - \frac{\eta}{2}\Big)\frac{1}{4}\log^2 Y \ge \Big(1 - \frac{\eta}{2}\Big)\Big(1 - O\Big(\frac{\log L}{L}\Big)\Big)\frac{L^2}{64} \ge (1 - \eta)\frac{(\log p)^2}{64}$$

for $X \ge X_0(\eta)$. The primes $p \le X^{1/2}$ number $\le \pi(X)/L$ for large X, and $O(\eta^{-2}XL^{-12}) \le \pi(X)/L$ for $X \ge X_0(\eta)$. So the exceptional set has at most $(1500\eta^{-2} + 2)\pi(X)/L \le 1600\eta^{-2}\pi(X)/L$ elements. Finally $f(p) \ge 3f_{\text{II}}(p)$ by [8, (1.4)] and $f_{\text{II}}(p) \ge R(p)$ by Lemma 2.1(iii). □

Remark 5.11. (1) The same proof applies within any fixed reduced residue class $b \bmod W$: all but $O_W(\eta^{-2}\pi(X;W,b)/\log X)$ primes $p \equiv b \pmod W$, $p \le X$, satisfy the bound, because $\pi(X;W,b) \gg_W \pi(X)$. This includes the six classes modulo 840 that contain all potential counterexamples.

(2) The constant $\frac{1}{64}$ is $2 \cdot \frac{1}{8} \cdot (\frac{1}{4})^2$: level $\frac{1}{4}$, the $u \leftrightarrow B$ symmetry, and the mean $\frac{1}{8}\log^2 Y$. The weights $\varphi(m)/m$ lose a further constant factor (heuristically $\pi^2/12$) relative to the unweighted count. We have not tried to remove this loss or to add the other fibrations of R.

(3) Only the solutions certified by moduli $\le X^{1/4-o(1)}$ are used, so $(\log p)^2/64$ is a lower bound for this "cheap" part of $R(p)$. The full second moment $\sum_{p\le X} R(p)^2$ is not estimated. The rate $\pi(X)/\log X$ comes from the bound $V \ll \log^3 Q$ against $M^2 \asymp \log^4 Q$; numerically V seems to be of order $\log^3 Q$ for this family (§6), so Chebyshev's inequality alone would not improve the rate, but we have no lower bound for V.

6. Numerical remarks

None of the theorems depends on computation. The following checks were made with the short programs `es-count-note.checks.c` and `es-count-note.checks.py` next to this file (the latter performs the exact exponent bookkeeping of Theorems 3.12 and 3.14), and serve only as sanity checks.

- *Zeros of R.* Enumerating all triples with $4uBC-u-B \le 10^7$, the primes $p \equiv 1 \pmod 4$ below 10^7 with $R(p) = 0$ are exactly the fourteen listed in §1, and $R(m^2) = 0$ for every odd square $m^2 \le 10^7$ (Lemma 2.3).
- *The genus main term.* Direct evaluation of (4) gives $M_{\mathrm{gen}}(409) = 3.99907$, $M_{\mathrm{gen}}(577) = 5.35239$, $M_{\mathrm{gen}}(8089) = 6.79081$ $(= 0.0839\,(\log p)^2)$, $M_{\mathrm{gen}}(1000033) = 27.737$ $(= 0.1453\,(\log p)^2)$ and $M_{\mathrm{gen}}(8813281) = 29.083$ $(= 0.1137\,(\log p)^2)$, and $M_{\mathrm{gen}}(n) = 0$ for $n = 31^2, 35^2, 99^2$. The ratios $M_{\mathrm{gen}}(p)/(\log p)^2$ are thus more than ten times the constant $2/(3\pi^4) = 0.00684$ of Theorem B. Note that $p = 8813281$ has $R(p) = 0$: its genus count is not small, and the vanishing comes from the characters of order ≥ 3. In earlier internal computations (65 slices, $u \in \{1, 5, 13, 21\}$, twenty primes from 409 to about 10^8), the formula of Proposition 4.10, with L-values computed numerically, agreed with the exact slice sums to within 0.13% for all eighteen slices of length $X_u \ge 10^5$; these involve the primes 712321, 1000033, 1134241, 1724209, 1726201, 5212561, 8813281, 10000169, 99999989 and 100000049, six of which have $R(p) = 0$. The errors decrease with X_u; short slices are not in the asymptotic range.
- *The variance of §5.* The exact model variance V (the right side of Lemma 5.6 minus M^2, computed by a double sum over $\mathcal{T}$ with the program `scripts/z53_pz_exactV_trial.c` of the accompanying repository) is $V = 2.0988$, 6.9439, 10.887 for $Q = 10^2, 10^3, 3 \cdot 10^3$, i.e. $V/\log^3 Q \approx 0.021$ in each case, while $M/\log^2 Q \approx 0.120$. So the bound of Proposition 5.10 has the right order but a constant about 70 times too large.

7. Concluding remarks

(1) The three main results concern the same count $R(p)$, whose average over all integers is of order $(\log n)^2$, whose average over primes is at least of that order (Theorem C), and whose positivity at every large prime would imply the Erdős–Straus conjecture for large primes. Each result is of a kind compatible with the odd-square obstruction (Lemma 2.3): an upper bound uniform in n (Theorem A), a lower bound for a main term that vanishes at squares (Theorem B), and an average over primes (Theorem C). None of them gives $R(p) \ge 1$ at a prescribed prime.

(2) Theorem B reduces the lower-bound problem for $R(p)$ to the remainder $X(p) = R(p) - M_{\mathrm{gen}}(p)$, the contribution of characters of order ≥ 3. On the subfamily of §4, Theorem 4.12 disposes of the part of X that involves only characters modulo $8u$ at density level. The rest of $R_{\mathfrak{F}} - M_{\mathfrak{F}}$ involves the exact divisibility $w \mid p + u$; we have no method for it.

(3) For the upper bound, the method stops near the cube; at the balanced block $u, B \asymp n^{1/4}$ Remark 3.16 indicates that a genuinely bilinear estimate would be needed. Even with the conjectural exponent pair the cube block gives only $n^{5/18+\varepsilon}$, so a bound of the shape $R(n) \ll n^{1/4+\varepsilon}$ would need new ideas in every block.

(4) For Theorem C, improving $\frac{1}{64}$ towards the heuristically predicted mean $\frac{3}{2\pi^2}(\log p)^2$ (the sum over $(u, B) \in \mathcal{P}(p)$ of the densities $1/\varphi(4uB)$), or proving the analogous bound $f_{\mathrm{II}}(p) \gg (\log p)^3$ for almost all p (the true order of f_{II} on average), would require control of correlations at larger level.

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